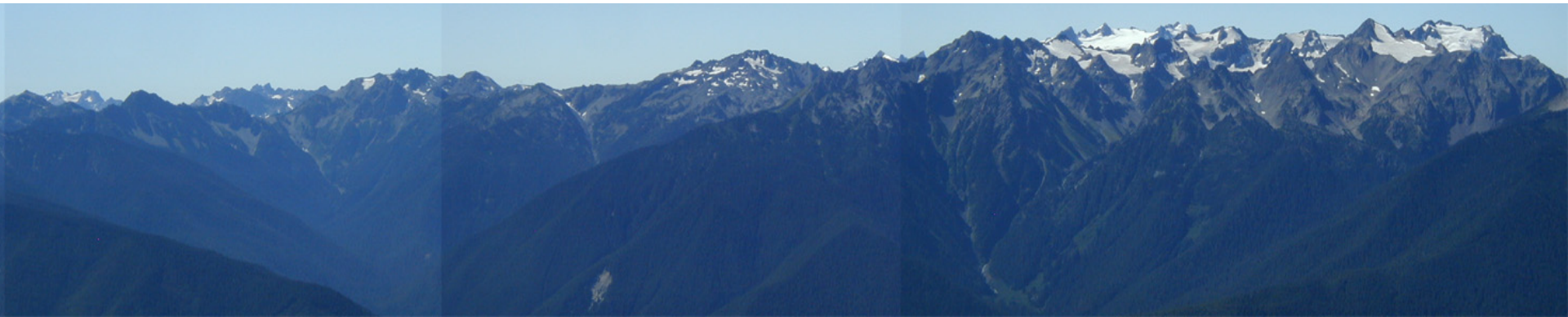
This Space Left Intentionally Blank

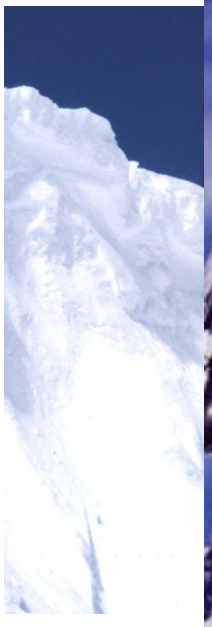
Quantum Error Correcting Subsystem Codes and Fault-Tolerant Quantum Computing

Dave Bacon
Department of Computer Science & Engineering
University of Washington, Seattle, WA

Funded by:



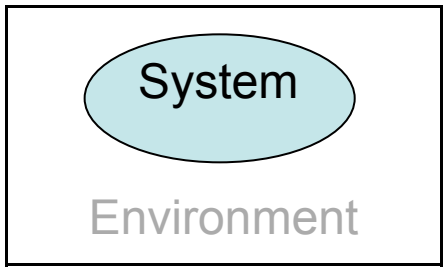
Paradise?





Problem Child

decoherence:



$$H = H_S \otimes I + I \otimes H_B + H_I$$

open quantum systems can become entangled with their environment

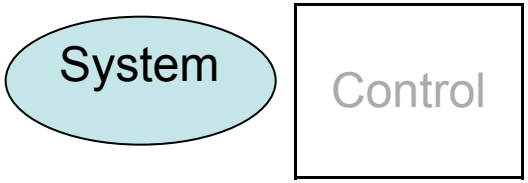
$$(\alpha|0\rangle_S + \beta|1\rangle_S) \otimes |0\rangle_E \rightarrow \alpha|0\rangle_S \otimes |0\rangle_E + \beta|1\rangle_S \otimes |1\rangle_E$$

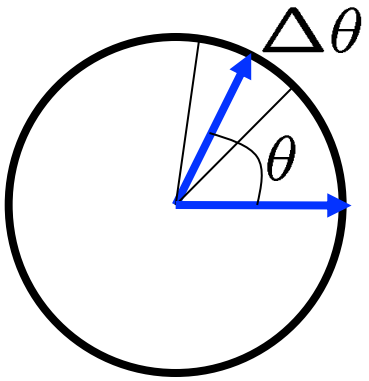
coherent superposition — incoherent sum

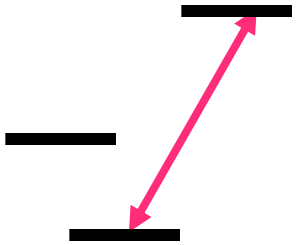
$$(\alpha|0\rangle_S + \beta|1\rangle_S) \rightarrow \rho = |\alpha|^2 |0\rangle_S\langle 0|_S + |\beta|^2 |1\rangle_S\langle 1|_S$$

imprecise and faulty control:

unitary gates, preparation, measurements



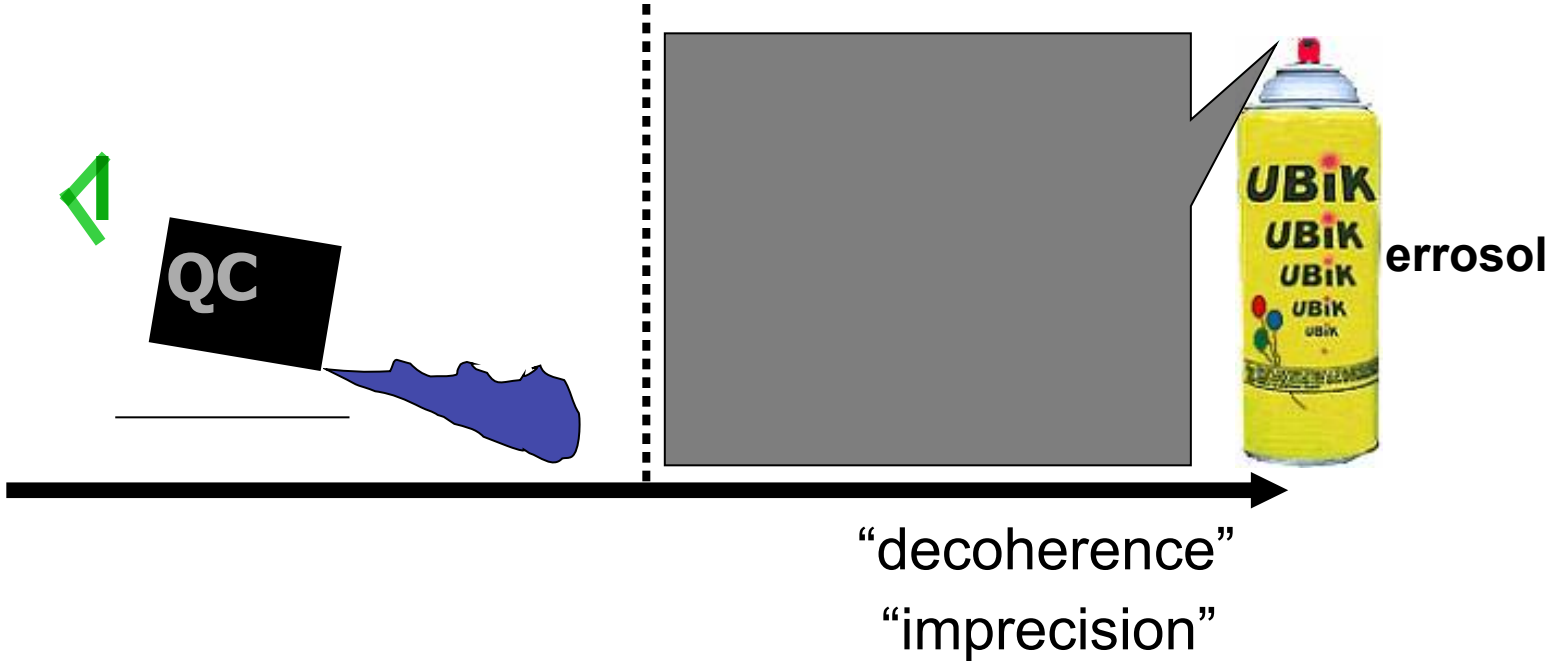




My Mortal Enemy



Problem Solution

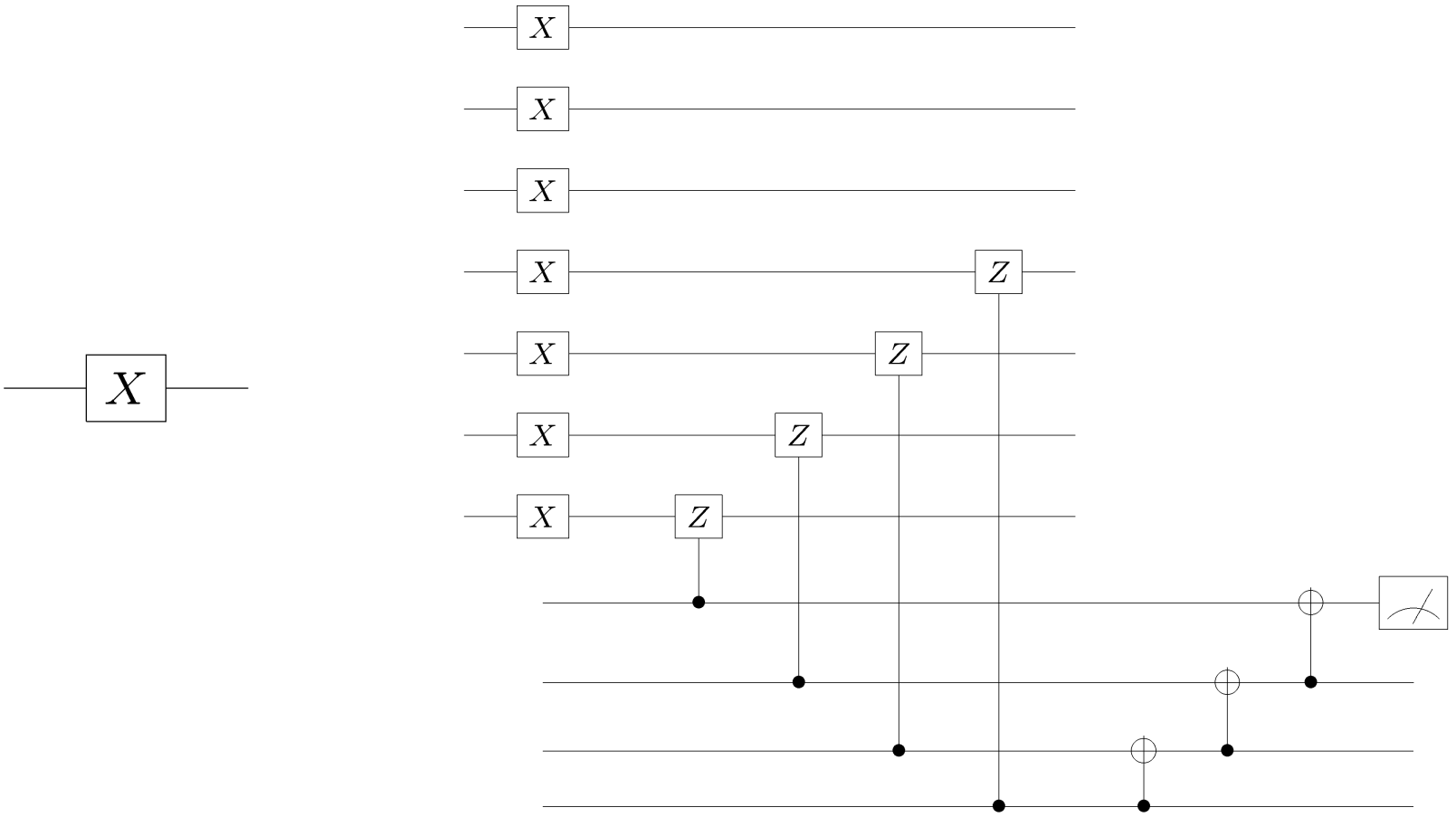


Threshold Theorem for Fault-Tolerant Quantum Computation

[Aharonov, Ben-Or, Knill, LaFlamme, Zurek, Shor, Gottesman, Preskill]

If decoherence is weak enough and control is precise enough, then a quantum computation to a desired accuracy can be carried out with an overhead polynomial in the log of the accuracy.

Sketch: How It Works



To make robust elements, information is encoded into a quantum error correcting codes of significant size and fault-tolerant methods are used for gates, preparation, measurement, and correction

In Practice?

How to build a 300 bit, 1 Gop quantum computer

Andrew M. Steane
Centre for Quantum Computation, Department of Atomic and Laser Physics, Clarendon Laboratory, Parks Road, Oxford, OX1 3PU, England
(Dated: December 22, 2005)

13000 ions, 1000 laser beams, 160000 electrodes in ion trap

The device complexity simply to engineer robust qubits is severe

“A quantum computer will spend most of its time doing error correction.”

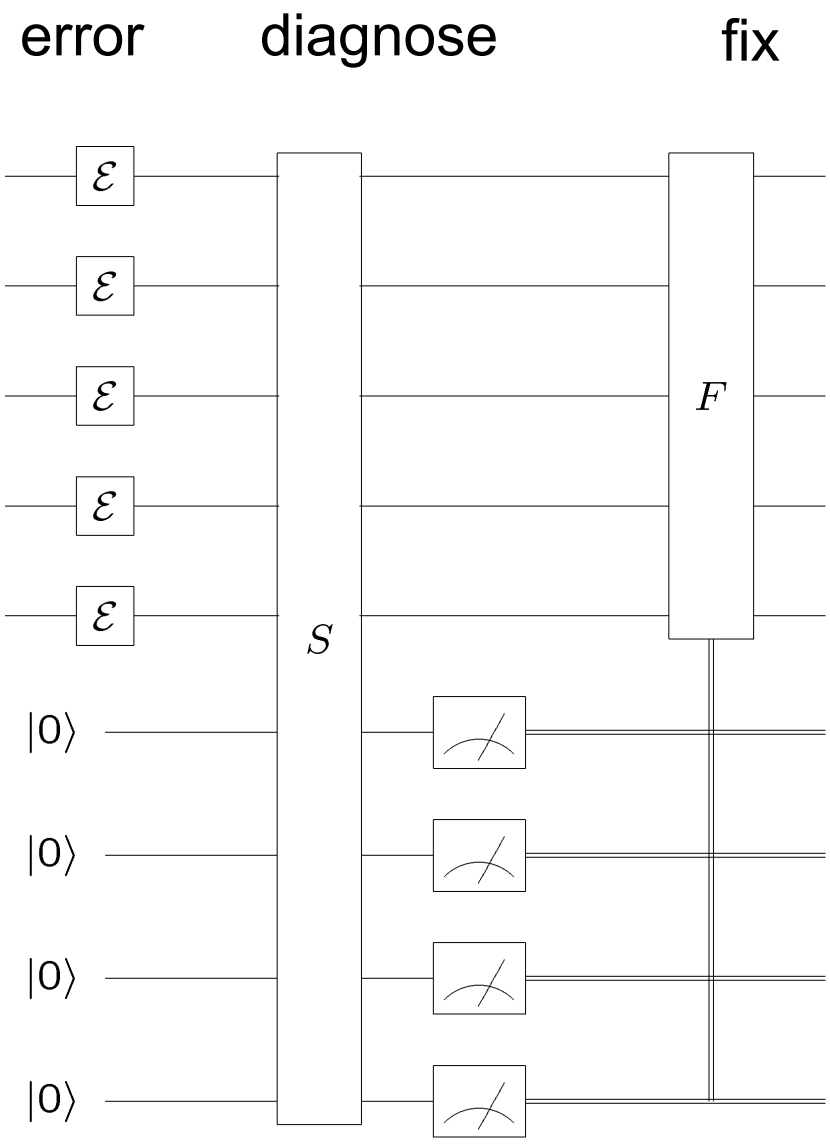
CONTRAST:





Is there a better way? Can we mimic classical computers?

Quantum Error Correction



encoded quantum information

encoded quantum information

"cold" ancilla qubits

"hot" ancilla qubits

Quantum error correction is essentially (sophisticated) cooling

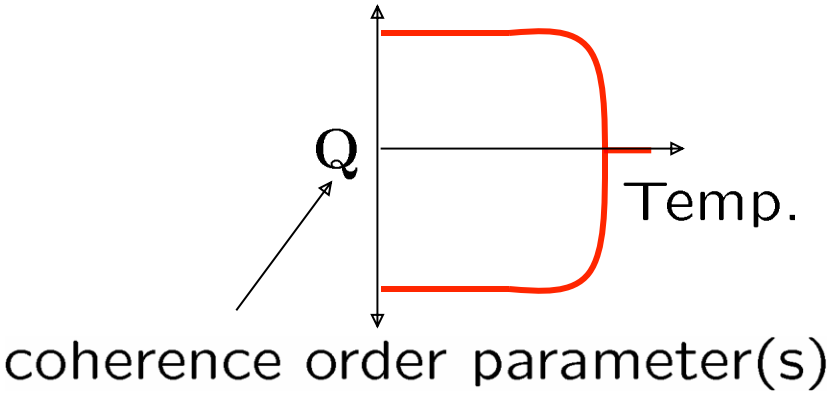
The Quantum Hard Drive?



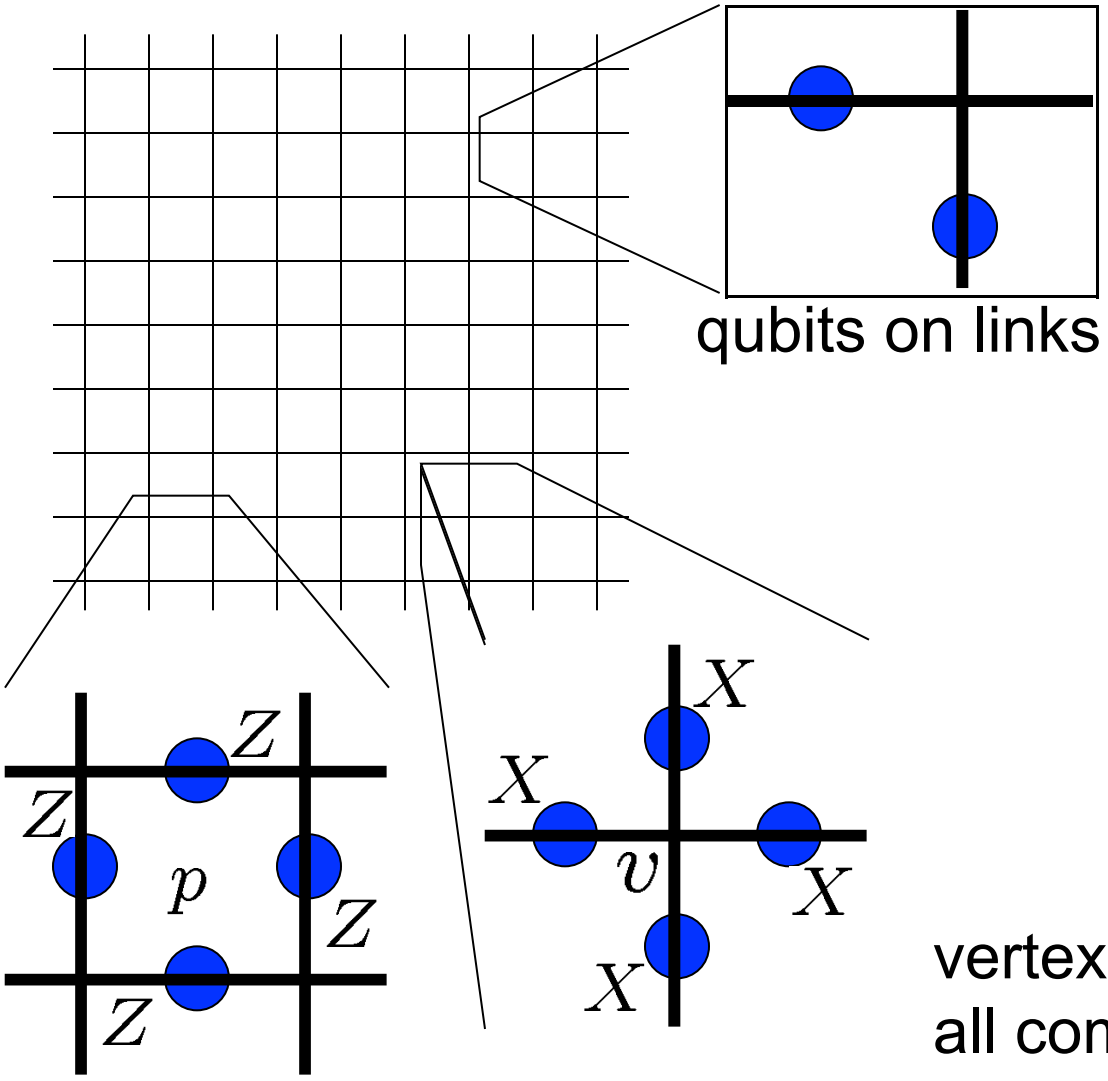
Kitaev



Do there exist (or can we engineer) quantum systems whose physics guarantees fault-tolerant quantum computation?



Kitaev's Model

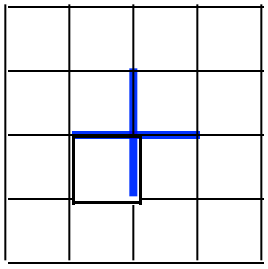


plaque operators:

$$S_p = \prod_{e \in p} Z_e$$

vertex operators:

$$S_v = \prod_{e \in v} X_e$$



vertex and plaque operators all commute with each other

$$Z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \quad X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

Kitaev's Model, Continued

Hamiltonian

$$H = -J\left(\sum_p S_p + \sum_v S_v\right)$$

$$S_p|\psi\rangle = s_p|\psi\rangle \quad s_p \in \{-1, +1\}$$

$$S_v|\psi\rangle = s_v|\psi\rangle \quad s_v \in \{-1, +1\}$$

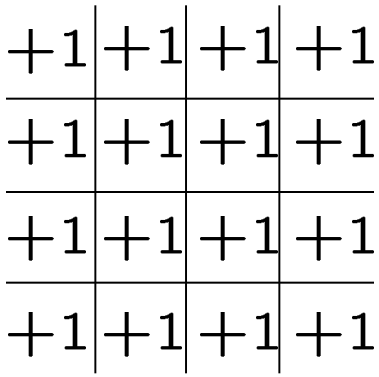
plaque operators:

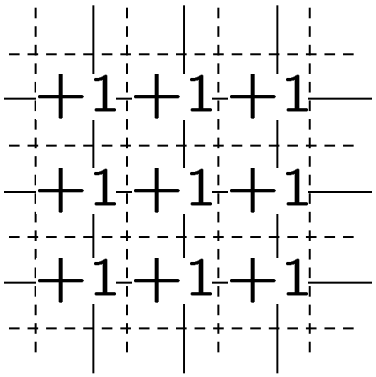
$$S_p = \prod_{e \in p} Z_e$$

vertex operators:

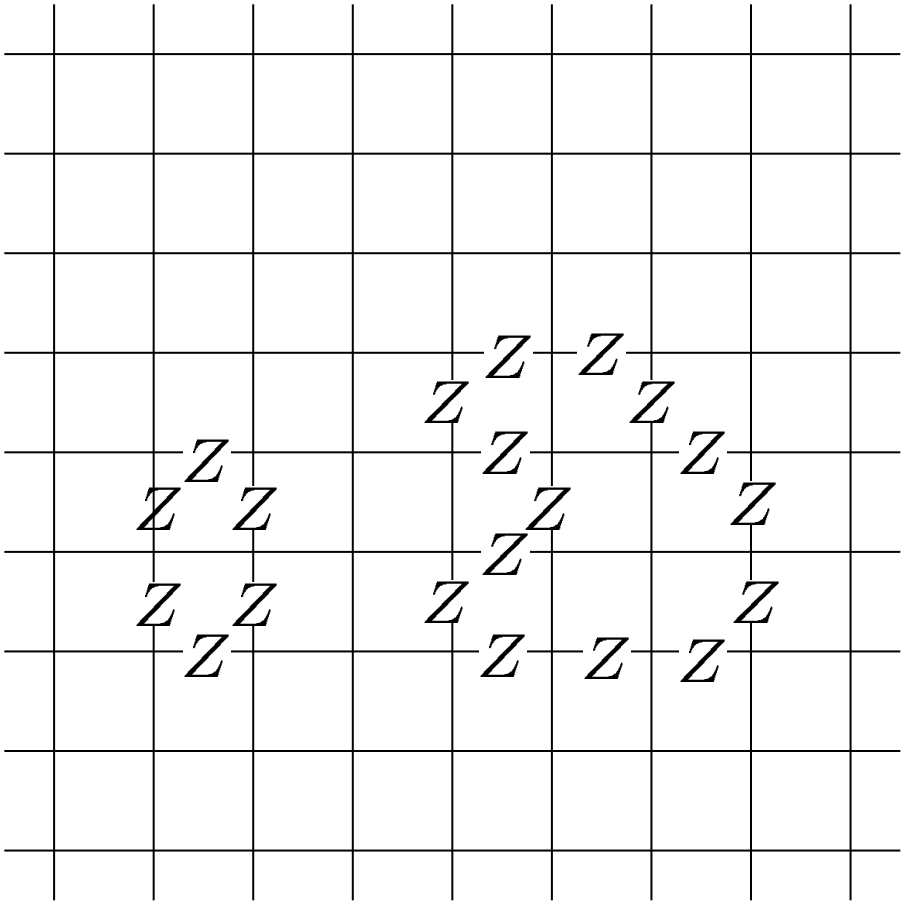
$$S_v = \prod_{e \in v} X_e$$

ground state has all $s_p = s_v = +1$





Topology of Operators


Z Z Z

Degeneracy

Hamiltonian

$$H = -J\left(\sum_p S_p + \sum_v S_v\right)$$

plaque operators:

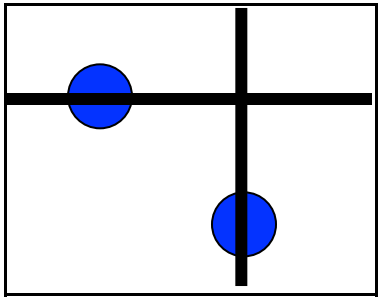
$$S_p = \prod_{e\in p} Z_e$$

vertex operators:

$$S_v = \prod_{e\in v} X_e$$

$2l^2$ qubits

$\dim\mathcal{H} = 2^{2l^2}$

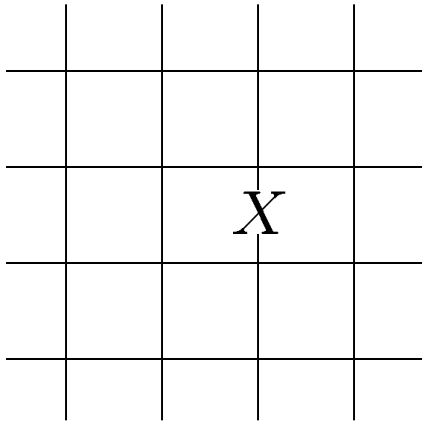


l^2 plaque operators $\prod_p S_p = I$ 2^{l^2-1} $s_p \in \{+1,-1\}$ values

l^2 vertex operators $\prod_v S_v = I$ 2^{l^2-1} $s_v \in \{+1,-1\}$ values

every energy level is $2^{2l^2} - 2^{l^2-1} - 2^{l^2-1} = 4$ degenerate

Kitaev's Model



plaque operators:

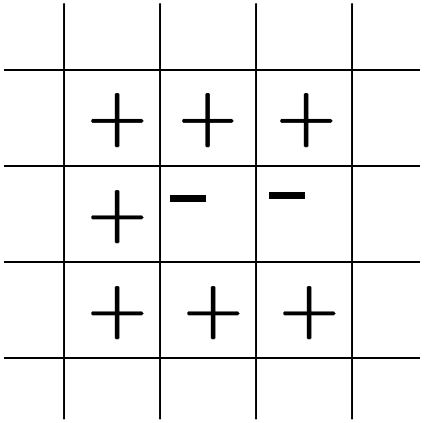
$$S_p = \prod_{e \in p} Z_e$$

vertex operators:

$$S_v = \prod_{e \in v} X_e$$

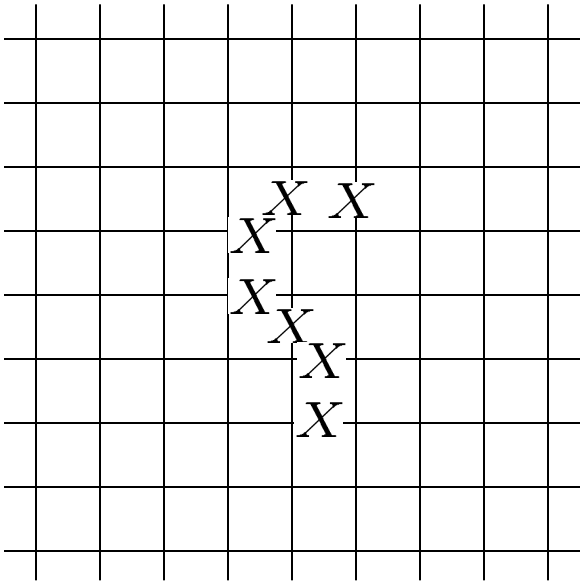
X anti-commutes with two plaque operators

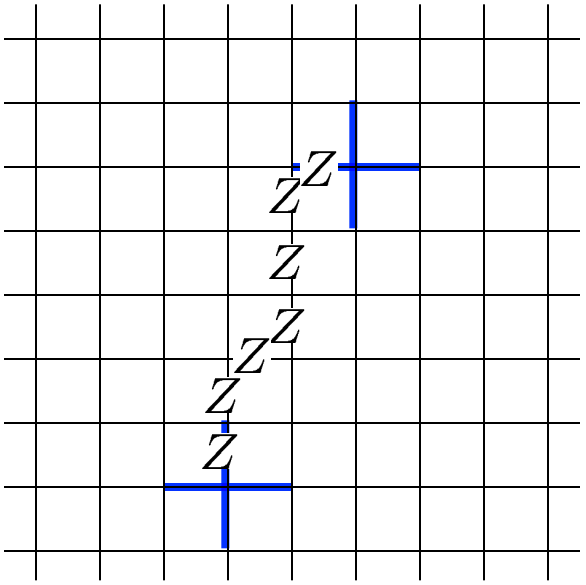
$$\{X, S_p\} = 0 \Rightarrow S_p(X|\psi_G\rangle) = -X|\psi_G\rangle$$



excitation is $4J$ above ground state

Excitations

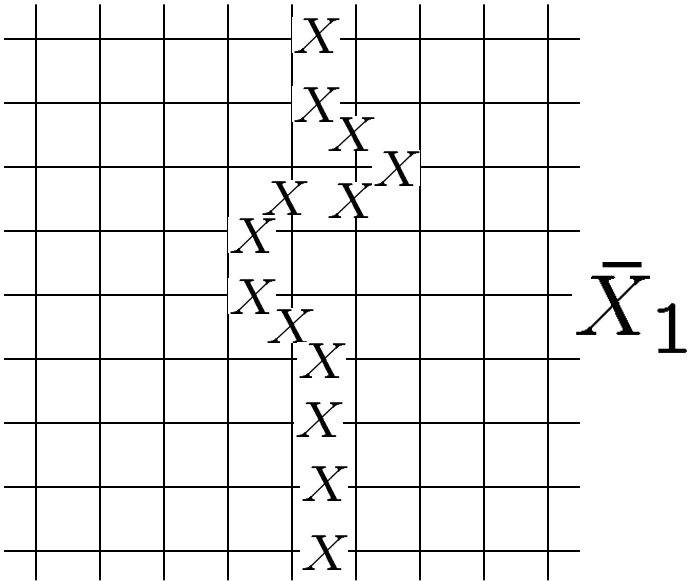


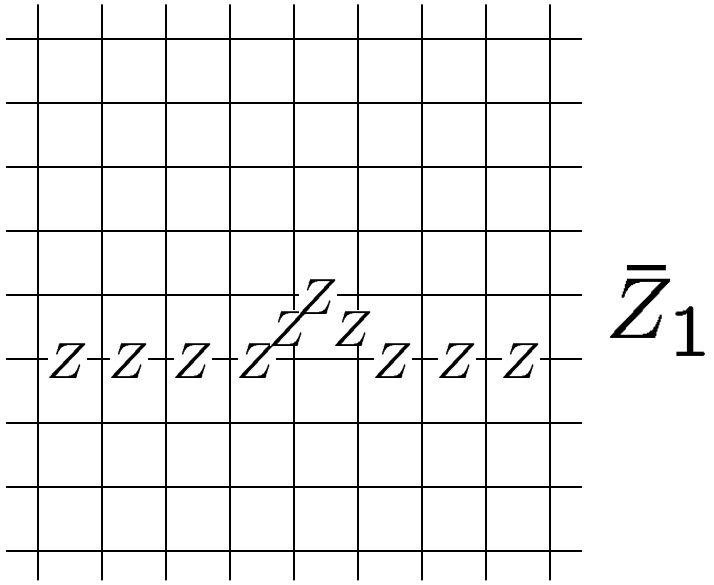


excitations particles come in pairs (particle/antiparticle)
at end of "error" chains

two types of particles, X-type (live on vertices of dual lattice)
Z-type (live on vertices of the lattice)

A Qubit



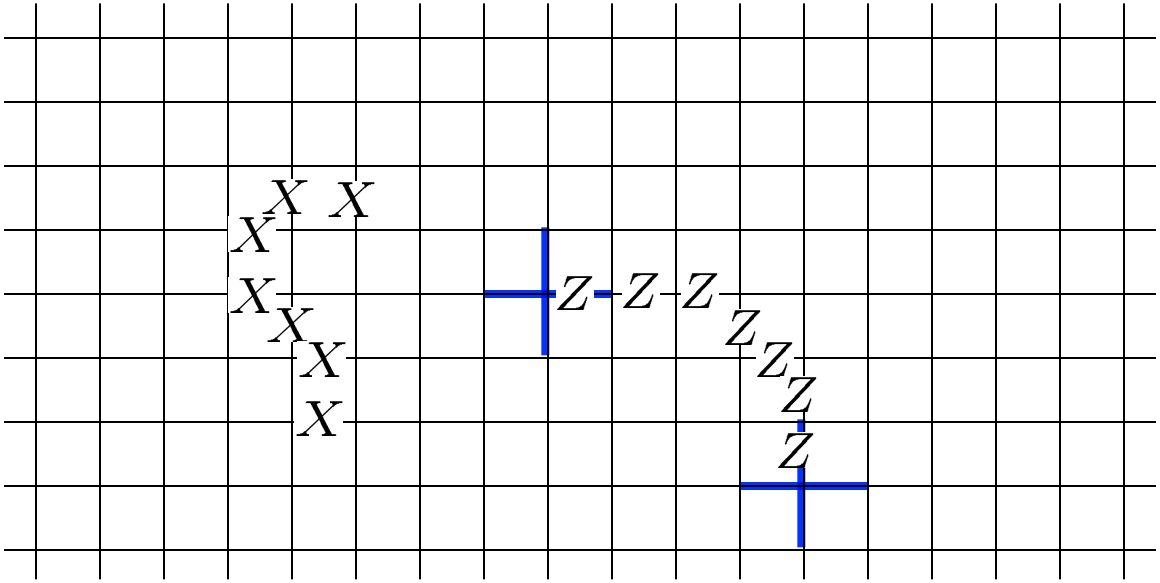


Moving an X or Z type particle on a homologically nontrivial path and annihilating with it's partner preserves the ground states

$$\bar{X}_1 \bar{Z}_1 \bar{X}_1 \bar{Z}_1 = -I$$

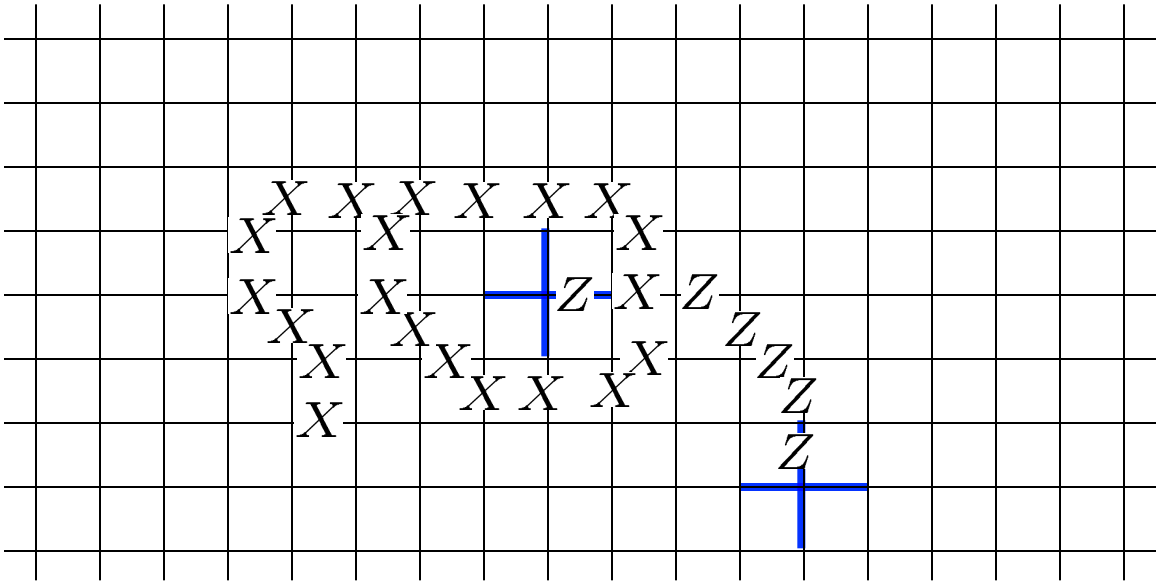
Action on ground state must be representation of non-Abelian group: Acts as encoded single qubit Pauli operator.

Anyons Anyone?



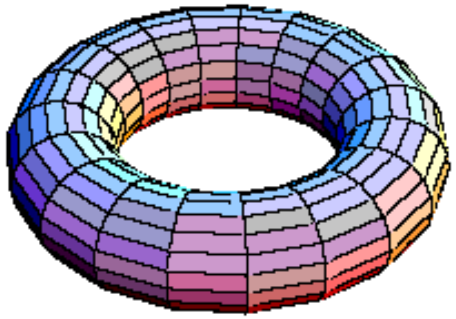
Phase:

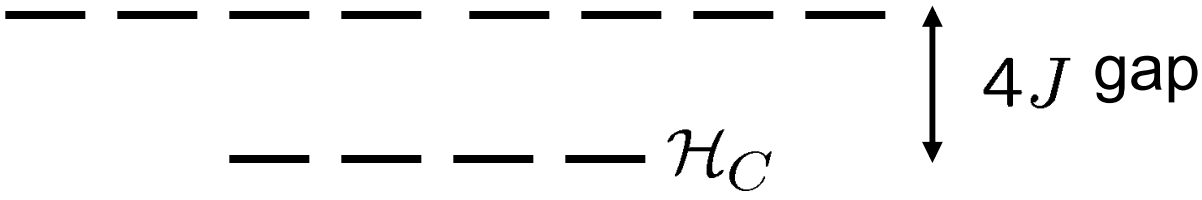
-1



Topological P[illegible]

Encode two qubits
into the ground state





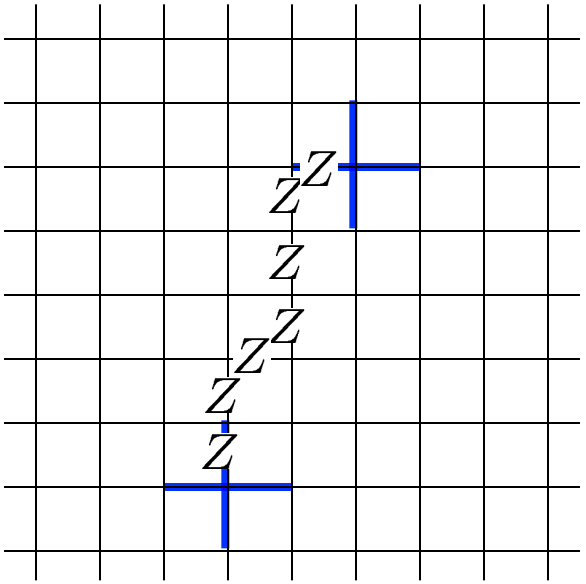
Perturbation theory:

$$V = \sum_v E_v + \sum_{vw} F_{vw}$$

But for $m \leq \left\lfloor \frac{l}{2} \right\rfloor$

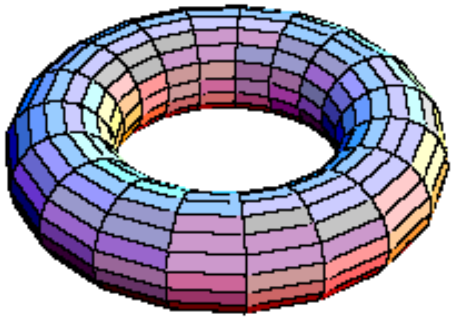
$$\langle \phi_i | V^m | \phi_j \rangle = 0$$

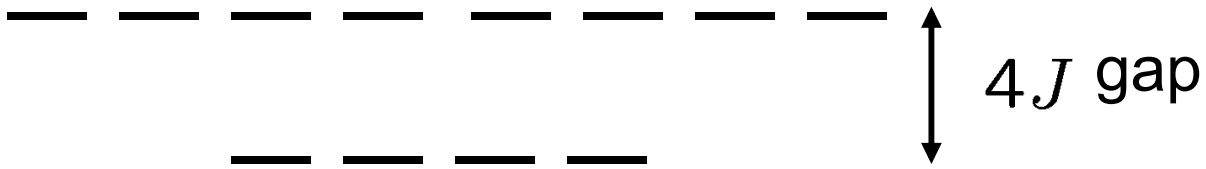
$$\langle \phi_i | V^m | \phi_i \rangle - \langle \phi_j | V^m | \phi_j \rangle = 0$$



Topological Quantum Computing

Encode two qubits
into the ground state





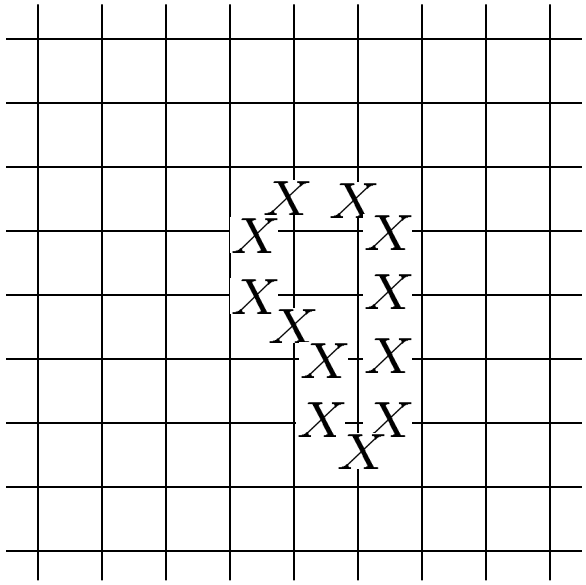
If we keep $kT \ll 4J$ then quantum information encoded into ground state will be “topologically” robust to perturbation.

We can perform gates (in this example X and Z) by creating anyon pairs and moving them around the torus.

Are these assumptions reasonable?

Topological Error Correction

Different mode of using Kitaev's Model, as a component in a fault-tolerant quantum computer, encode into ground state



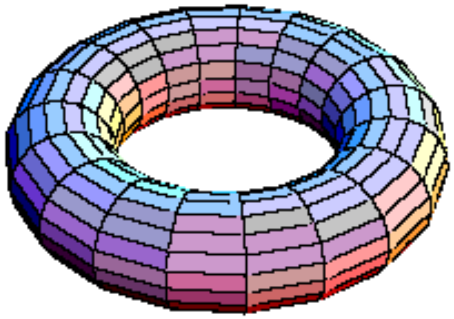
1. Errors occur
2. Measure vertex and plaque operators
3. Apply recovery operator
4. If created loop is homologically trivial, then we have restored information encoded into ground state.

Full analysis: measurements uncertain, recovery may fail, etc.
Threshold for fault-tolerant storage of quantum information:

[Dennis, Kitaev, Landahl, Preskill 2002]
[Wang, Harrington, Preskill 2003]

Fault-Tolerant?

Encode two qubits
into the ground state



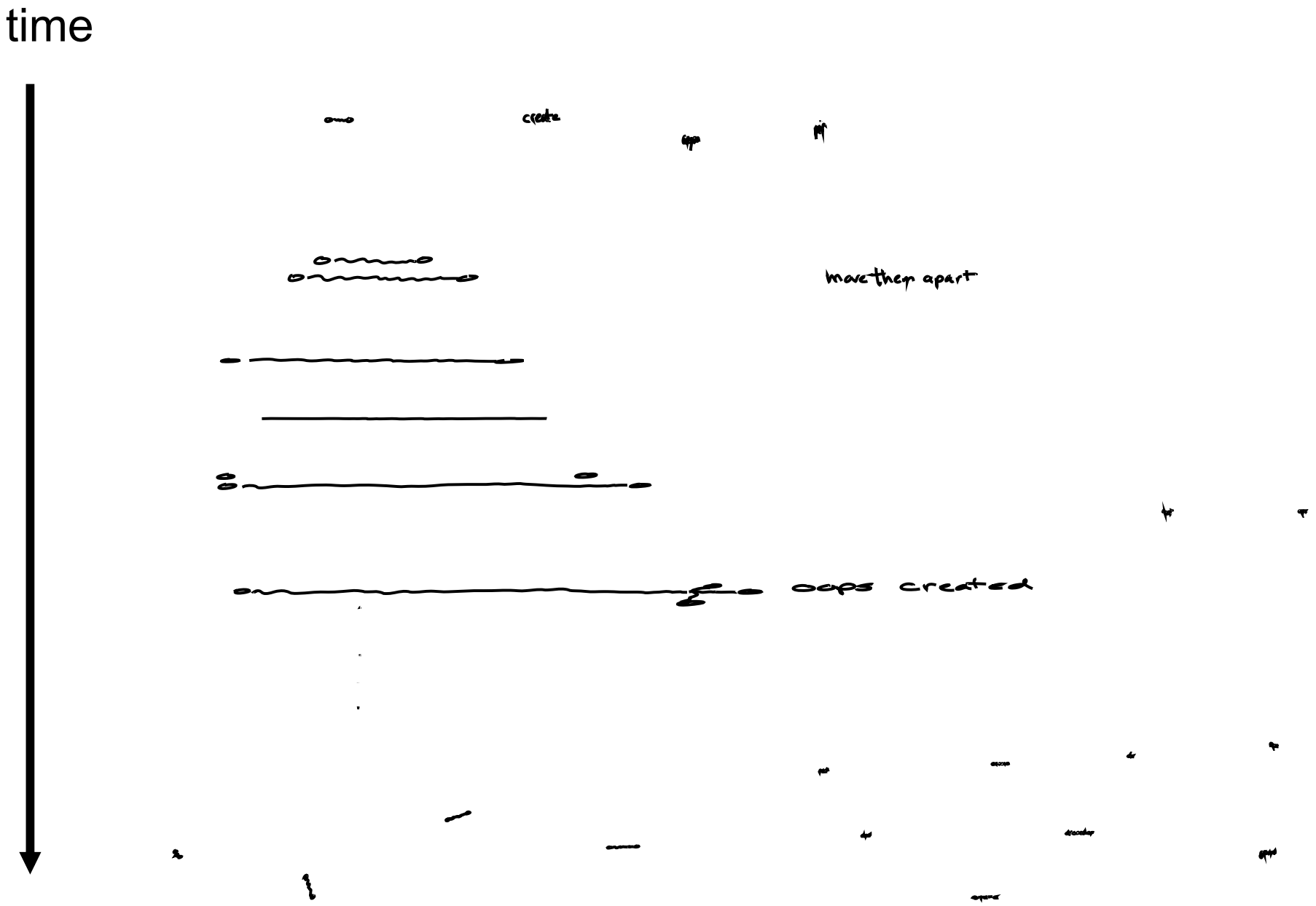


If we keep $kT \ll 4J$ then quantum information encoded into ground state will be “topologically” robust to perturbation.

We can perform gates (in this example X and Z) by creating anyon pairs and moving them around the torus.

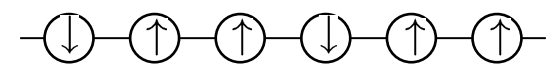
Are these assumptions reasonable?

Fault-Tolerant?

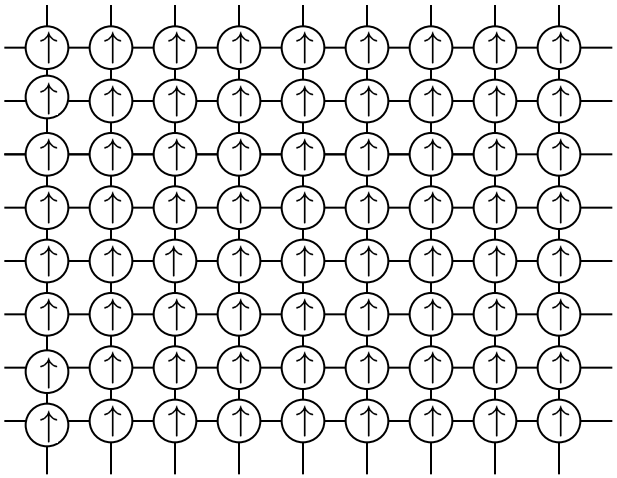


Three Storage Testbeds

Ferromagnetic Ising models:



one dimensional



two (and >) dimensional

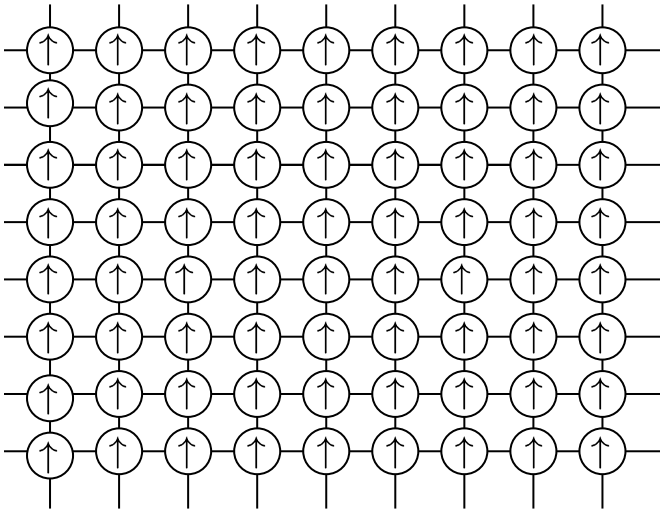
Toom's rule:

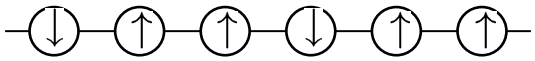
0	0	1	0	0	0	0	0
0	1	1	0	1	0	0	0
0	1	0	0	0	0	0	0
0	0	1	1	1	1	1	0
0	0	0	0	1	0	1	0
1	1	1	0	0	0	0	0

two dimensional noisy cellular automata

Ferromagnetic Ising Models







Ising model

spins on edges of a lattice: $s_i \in \{+1, -1\}$

energy of a configuration:

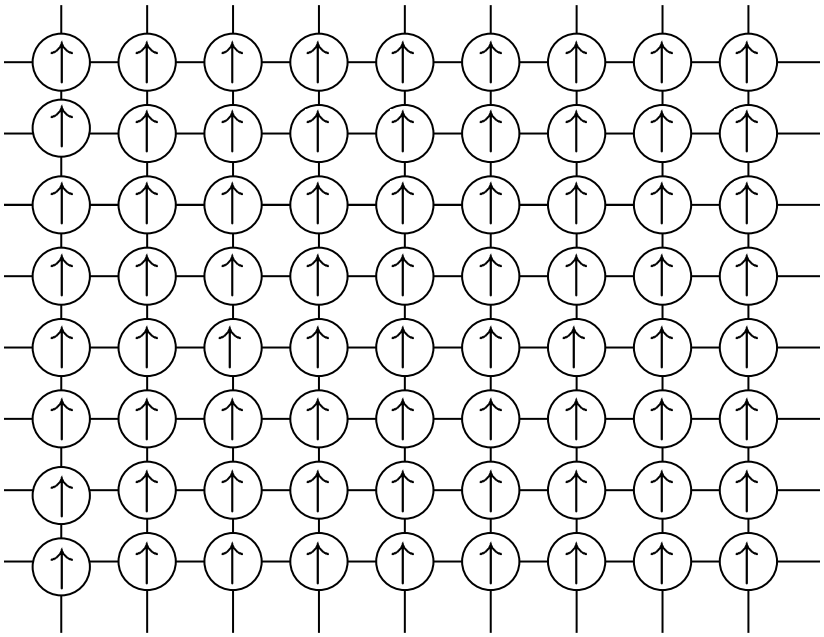
$$E = -J \sum_{\langle ij \rangle} s_i s_j + h \sum_i s_i \qquad J > 0$$

sum over neighbors

external magnetic field

$h = 0$ for now

Dynamic Ising Models



Dynamic Ising model (Metropolis update)

Interaction with environment causes spin flips

Update rule:

If flipping spin would decrease energy, flip spin

If flipping spin would increase energy, flip spin with probability

$\exp(-\beta \Delta E)$



Reminder



two-level system perturbatively coupled to a thermal boson bath

$$\frac{\partial \boldsymbol{\rho}}{\partial t} = -\bar{n}_{th}\frac{\gamma}{2}\left[\boldsymbol{\sigma}_-\boldsymbol{\sigma}_+\boldsymbol{\rho} + \boldsymbol{\rho}\boldsymbol{\sigma}_-\boldsymbol{\sigma}_+ - 2\boldsymbol{\sigma}_+\boldsymbol{\rho}\boldsymbol{\sigma}_-\right]$$

"heating"

$$-(\bar{n}_{th}+1)\frac{\gamma}{2}\left[\boldsymbol{\sigma}_+\boldsymbol{\sigma}_-\boldsymbol{\rho} + \boldsymbol{\rho}\boldsymbol{\sigma}_+\boldsymbol{\sigma}_- - 2\boldsymbol{\sigma}_-\boldsymbol{\rho}\boldsymbol{\sigma}_+\right]$$

"cooling"

$$-\phi\left[\boldsymbol{\rho} - \boldsymbol{\sigma}_z\boldsymbol{\rho}\boldsymbol{\sigma}_z\right]$$

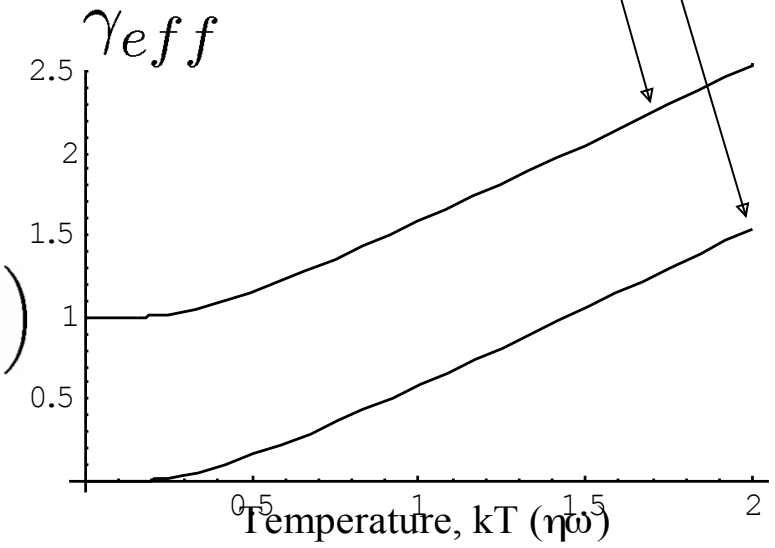
"non-dissipative"

$\hbar\omega$

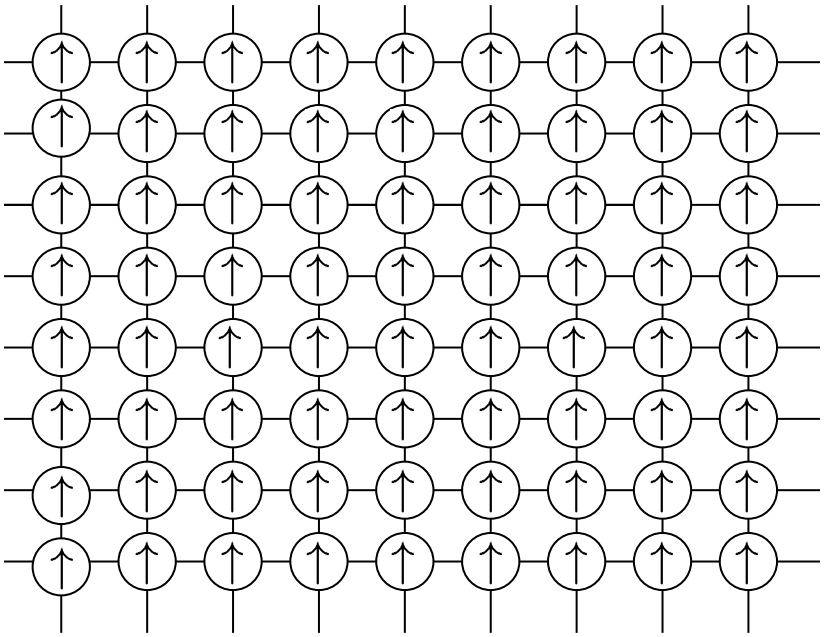
$$\bar{n}_{th} = \frac{1}{\exp\left(\frac{\hbar\omega}{kT}\right)+1}$$

$$\frac{\text{heating}}{\text{cooling}} = \frac{\bar{n}_{th}}{\bar{n}_{th}+1} = \exp\left(-\frac{\hbar\omega}{kT}\right)$$

$$= \exp\left(-\beta\Delta E\right)$$



Dynamic Ising Models



Dynamic Ising model (Metropolis update)

Interaction with environment causes spin flips

Update rule (alternate with noise):

If flipping spin would decrease energy, flip spin

If flipping spin would increase energy, flip spin with probability $\exp(-\beta \Delta E)$

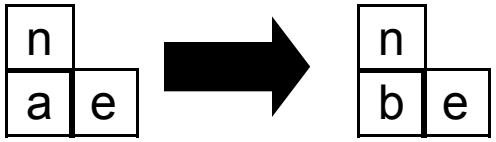
Toom's Rule



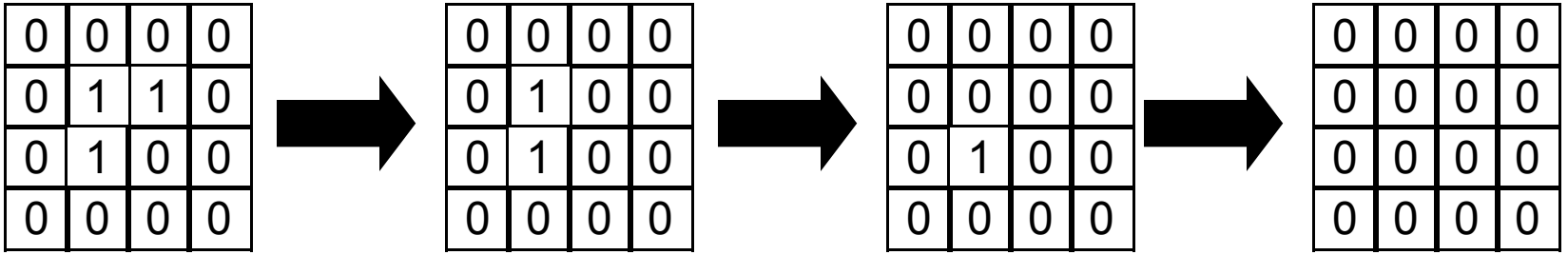
0	0	1	0	0	0	0	0
0	1	1	0	1	0	0	0
0	1	0	0	0	0	0	0
0	0	1	1	1	1	1	0
0	0	0	0	1	0	1	0
1	1	1	0	0	0	0	0

two dimensional noisy cellular automata

Update rule:



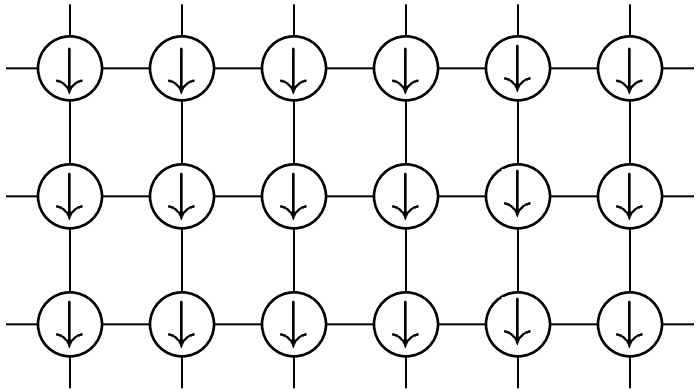
b=majority(n,e,b)



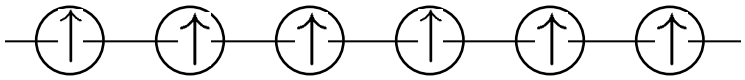
Update rule with noise: with some probability the update fails and a random state is assigned

Compare and Contrast

Encode information into the system and watch what happens



0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0



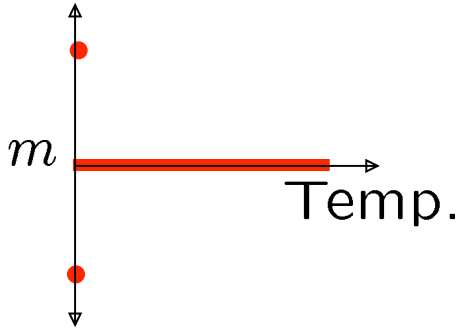
Redundancy code.

Order parameter is magnetization: $m = \frac{1}{N} \sum_i s_i$

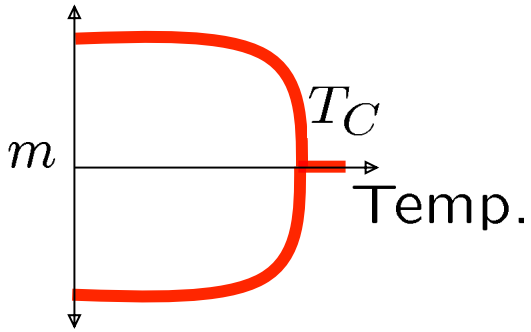
Storage

Long time behavior (not infinite time, but effectively)

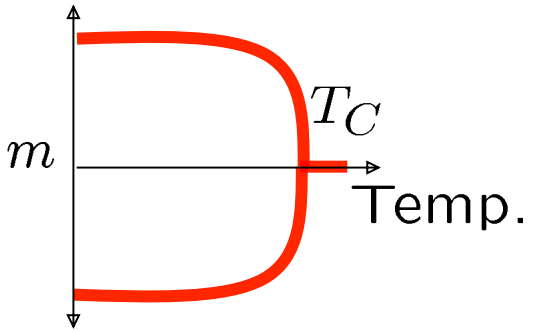
1D Ising



2D Ising



Toom's Rule



Criticism: (1) thermodynamic limit taken

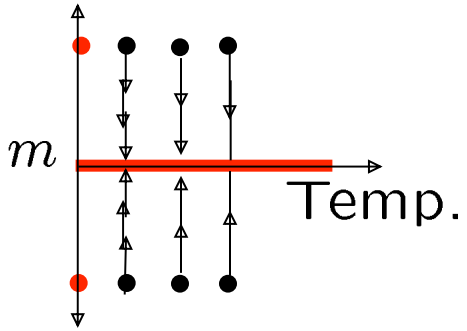
(2) what if relaxation to equilibrium takes a long time?

Storage



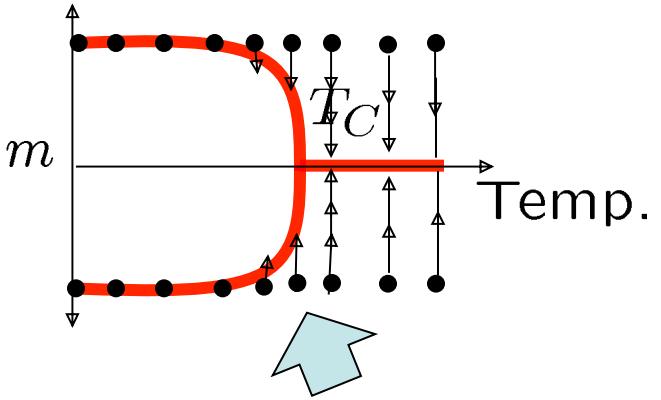
2nd attempt:
1D Ising

Relaxation to thermal



decay rate $\propto e^{-\beta J}$

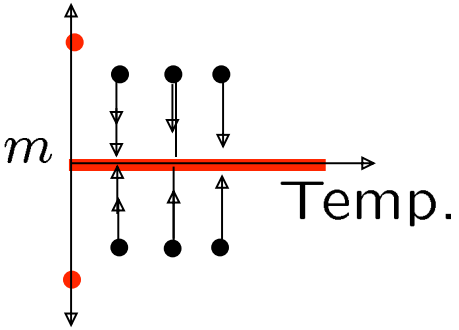
2D Ising, Toom's Rule

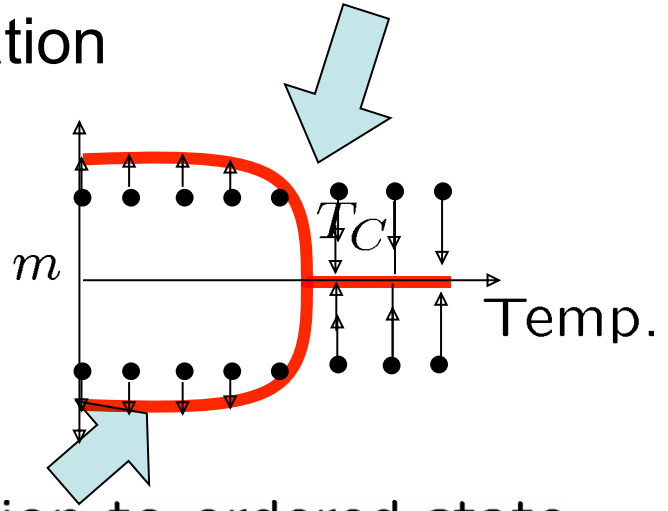


critical slowdown

$t_{th} \propto |T - T_c|^{z-\nu}$

Imperfect preparation





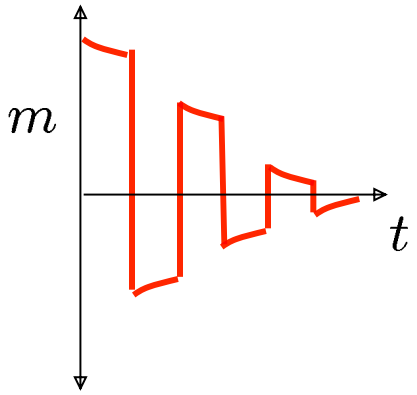
fast relaxation to ordered state

Manipulation



Flipping spins is imperfect

1D Ising

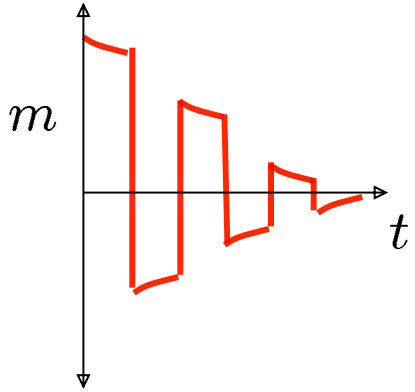


1D Ising
- temperature less than gap
- manipulation error must be small

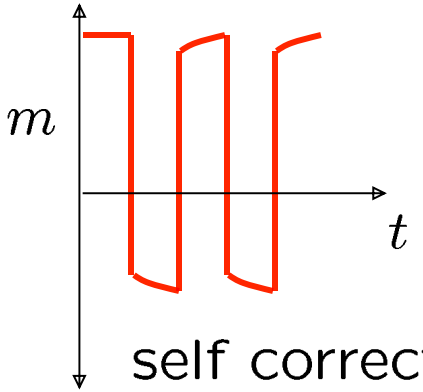
(i.e. NOT fault-tolerant)

2D Ising
- temperature less than critical temp

2D Ising
Toom's Rule



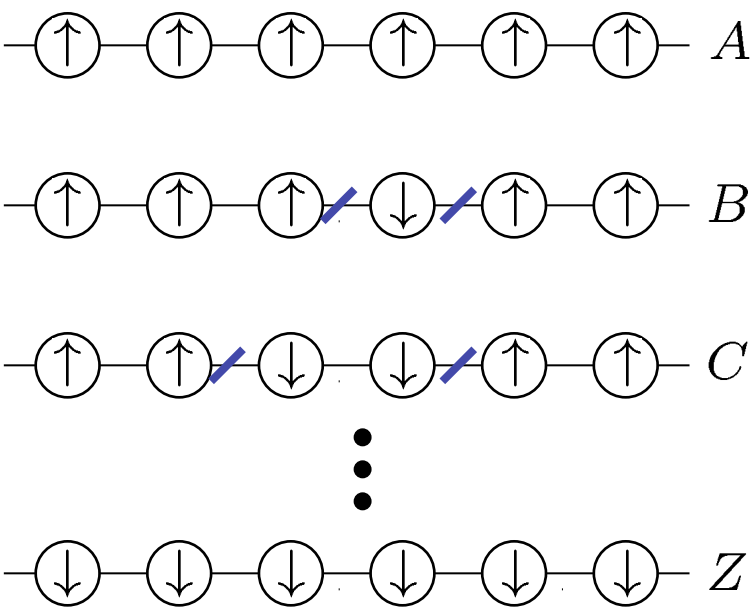
$T > T_C$

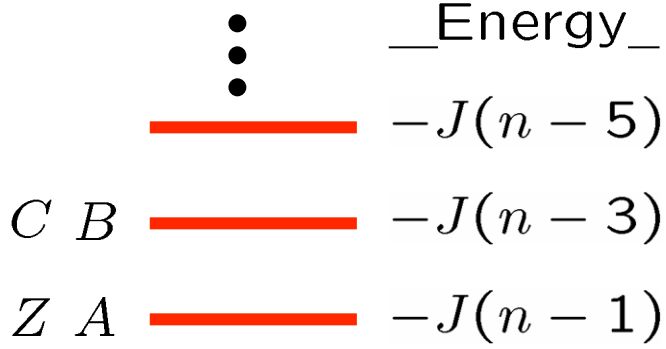


$T < T_C$

self correcting

“Self Fixing”

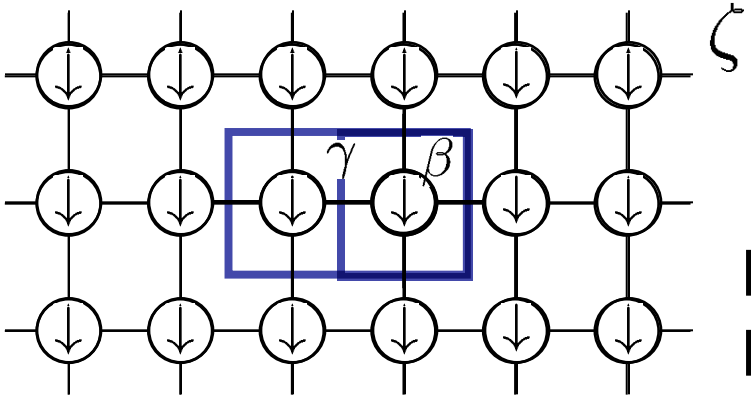


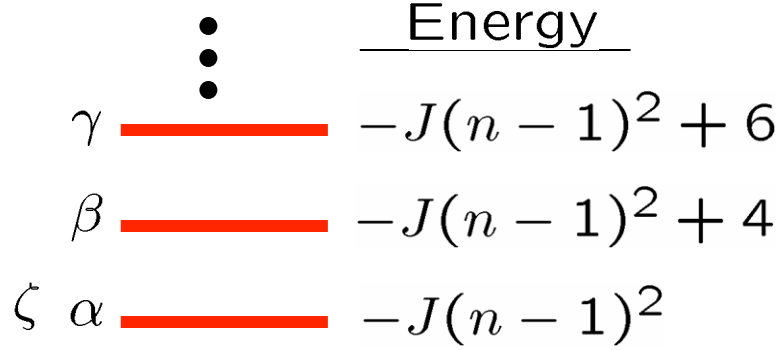


Excitations point-like

Energy to err is constant

Disordering entropy >> Energy



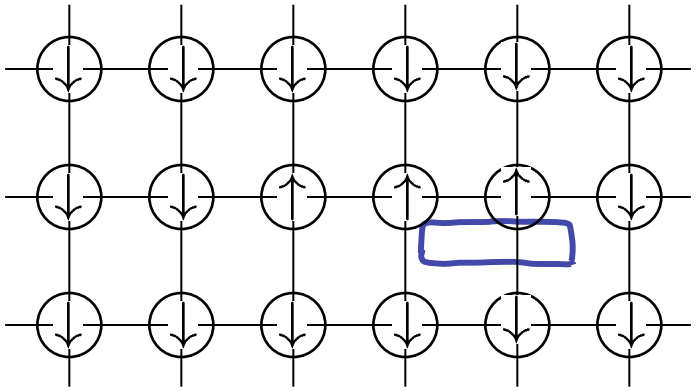


Excitations string-like

Energy to error is prop to perimeter

Energy >> Disordering entropy

Self-Correction

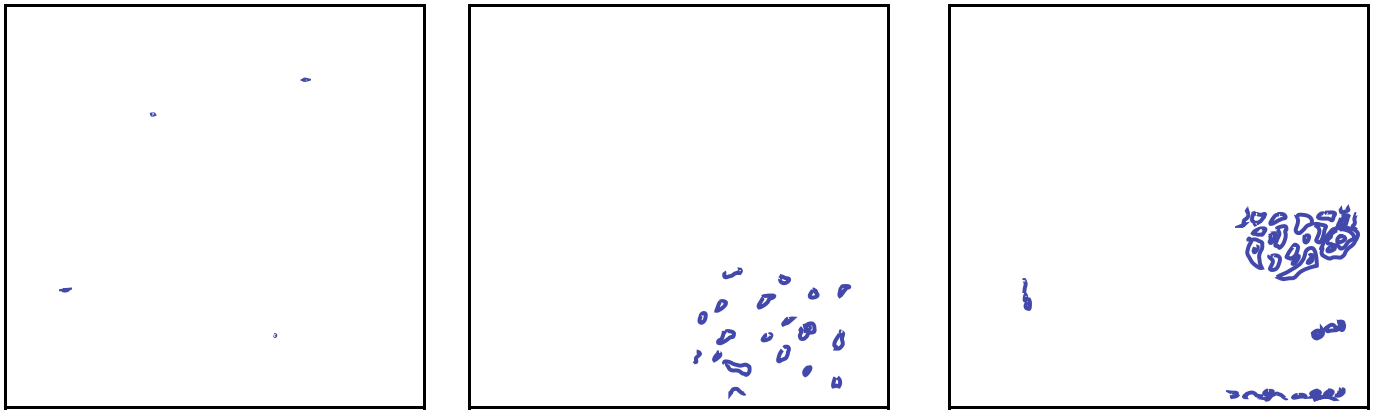


Energy:

$$E = -J \sum_{\langle ij \rangle} s_i s_j \qquad J > 0$$

sum over neighbors

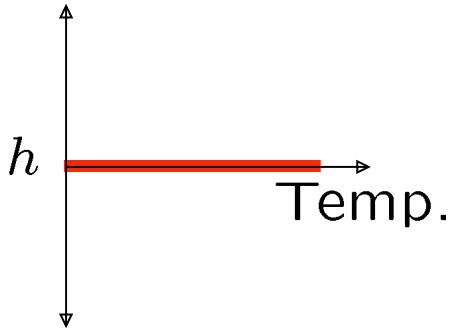
Energy is proportional to perimeter



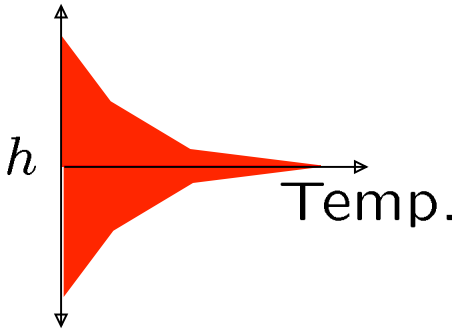
Toom's Rule Versus 2D Ising

Applied magnetic field (corresponds to a bias in noise)

2D Ising



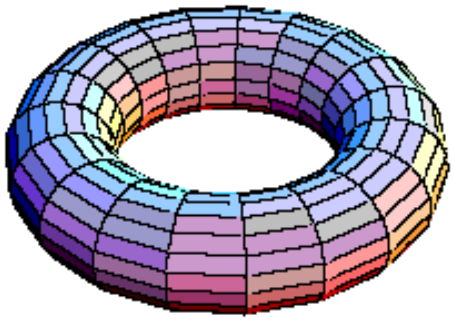
Toom's Rule



Regions with two coexisting phases

External magnetic fields disorder magnetic media

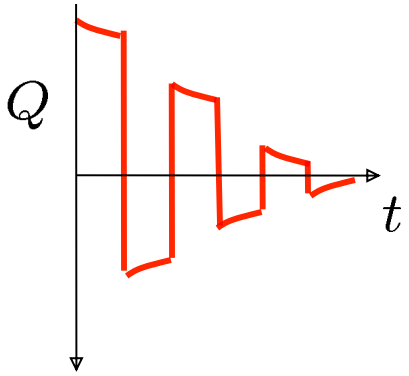
Whence Kitaev's Model?

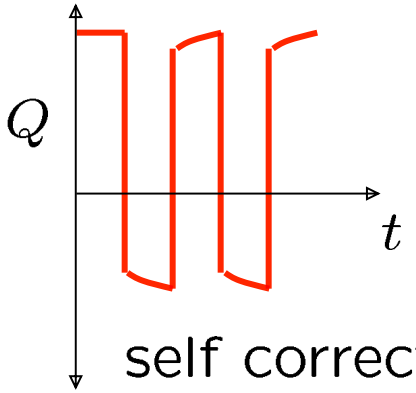


$$H = -J\left(\sum_p S_p + \sum_v S_v\right)$$

obust storage of quantum information.

ut it does not provide a mechanism for
bust manipulation of quantum information.



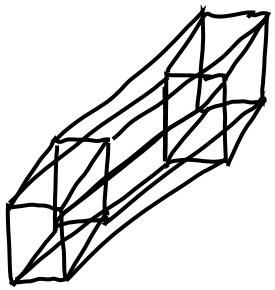


$T < T_C$

self correcting

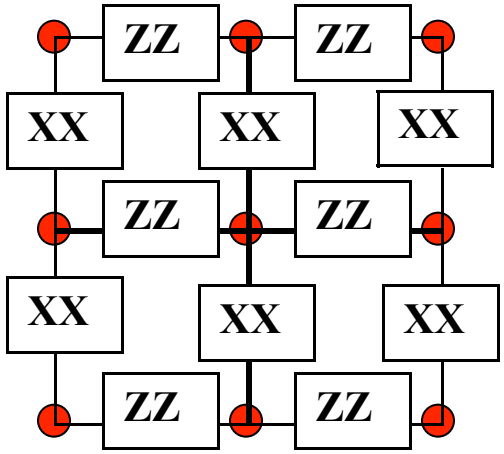
Is it possible to construct a Hamiltonian, which allows for fault-tolerant manipulation as well?

Into the 4th Dimension



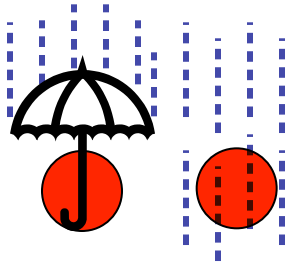
Quantum Compass Model

$$H = -J_X \sum_{ij} X_{i,j} X_{i+1,j} - J_Z \sum_{ij} Z_{i,j} Z_{i,j+1}$$



Ground state is a new kind of quantum error correcting code:
quantum error correcting subsystem

Subsystem Error Correction



Only need to protect subsystem, not full subspaces.

$$\mathcal{H}_{sys} = (\mathcal{C} \otimes \mathcal{D}) \oplus \mathcal{P}$$

encoding subsystem

$(|\phi\rangle\langle\phi| \otimes \rho_D) \oplus 0$ — $\mathcal{E}$ — $\mathcal{R}$ — $(|\phi\rangle\langle\phi| \otimes \tilde{\rho}_D) \oplus 0$

error recovery

encoded quantum information

recovery need not perfectly restore system, but only preserve information encoded into subsystem.

Shor Stabilizer

Quantum version of parity check matrix

$$|0_L\rangle = \frac{1}{2\sqrt{2}}(|000\rangle+|111\rangle)\otimes(|000\rangle+|111\rangle)\otimes(|000\rangle+|111\rangle)$$

$$|1_L\rangle = \frac{1}{2\sqrt{2}}(|000\rangle-|111\rangle)\otimes(|000\rangle-|111\rangle)\otimes(|000\rangle-|111\rangle)$$

Stabilizer generators:

$$ZZIIIIIII \quad IIIZZIIII \quad IIIIIIZZI \quad XXXXXXIII$$
$$IZZIIIIII \quad IIIIZZIII \quad IIIIIIIZZ \quad IIIXXXXXX$$

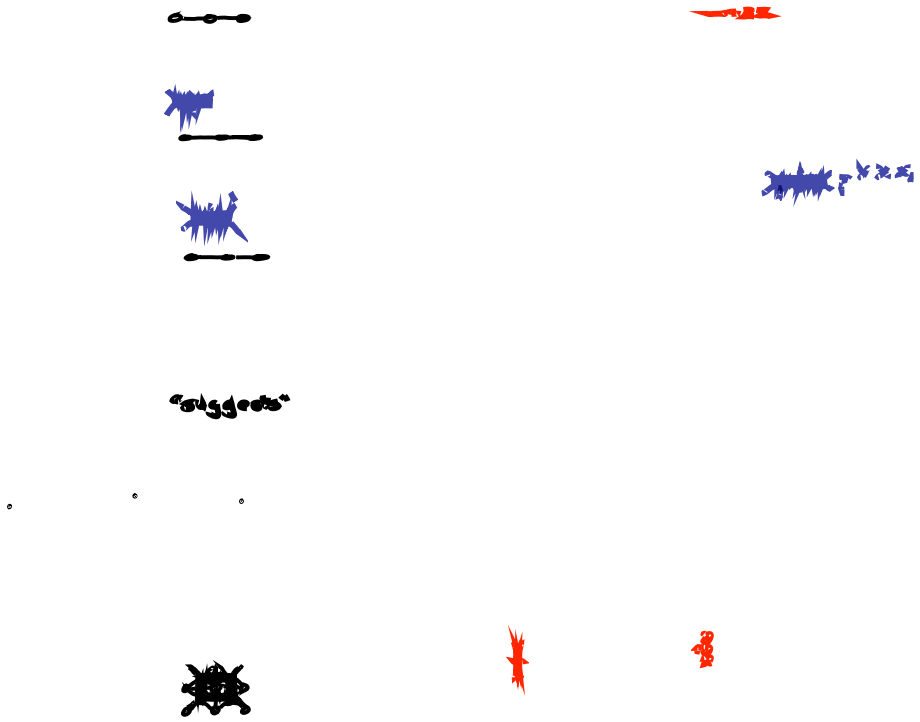
Define subspace:

$$S|\psi\rangle = |\psi\rangle$$

Logical operators:

$$\bar{Z} = ZZZZZZZZZ \qquad \bar{Z}|0_L\rangle = |0\rangle$$
$$\bar{X} = XXXXXXXXX \qquad \bar{Z}|1_L\rangle = -|1\rangle$$

$$ZZIIIIII \quad IIIZZIII \quad IIIIIIZZ \quad XXXXXXII$$
$$IZZIIIII \quad IIIIZZII \quad IIIIIIZZ \quad IIIXXXXX$$



$$\begin{matrix} X & X & X \\ X & X & X \\ I & I & I \end{matrix} \qquad \begin{matrix} I & I & I \\ X & X & X \\ X & X & X \end{matrix} \qquad \begin{matrix} Z & Z & I \\ Z & Z & I \\ Z & Z & I \end{matrix} \qquad \begin{matrix} I & Z & Z \\ I & Z & Z \\ I & Z & Z \end{matrix}$$

Shor Redux

Stabilizer generators:

$$\begin{matrix} X & X & X \\ X & X & X \\ I & I & I \end{matrix} \qquad \begin{matrix} I & I & I \\ X & X & X \\ X & X & X \end{matrix} \qquad \begin{matrix} Z & Z & I \\ Z & Z & I \\ Z & Z & I \end{matrix} \qquad \begin{matrix} I & Z & Z \\ I & Z & Z \\ I & Z & Z \end{matrix}$$

Encodes 9-4=5 qubits of information

Logical operators:

$$\bar{X}_1 = \begin{matrix} X & I & I \\ X & I & I \\ I & I & I \end{matrix} \quad \bar{X}_2 = \begin{matrix} I & I & X \\ I & I & X \\ I & I & I \end{matrix} \quad \bar{X}_3 = \begin{matrix} I & I & I \\ X & I & I \\ X & I & I \end{matrix} \quad \bar{X}_4 = \begin{matrix} I & I & I \\ I & I & X \\ I & I & X \end{matrix}$$

$$\bar{Z}_1 = \begin{matrix} Z & Z & I \\ I & I & I \\ I & I & I \end{matrix} \quad \bar{Z}_2 = \begin{matrix} I & Z & Z \\ I & I & I \\ I & I & I \end{matrix} \quad \bar{Z}_3 = \begin{matrix} I & I & I \\ I & I & I \\ Z & Z & I \end{matrix} \quad \bar{Z}_4 = \begin{matrix} I & I & I \\ I & I & I \\ I & Z & Z \end{matrix}$$

$$\bar{X}_5 = \begin{matrix} X & X & X \\ X & X & X \\ X & X & X \end{matrix} \quad \bar{Z}_5 = \begin{matrix} Z & Z & Z \\ Z & Z & Z \\ Z & Z & Z \end{matrix}$$

[Bacon, Phys. Rev A, 73, 012340]

Shor Redux

$$\begin{matrix} X & X & X \\ X & X & X \\ I & I & I \end{matrix} \qquad \begin{matrix} I & I & I \\ X & X & X \\ X & X & X \end{matrix} \qquad \begin{matrix} Z & Z & I \\ Z & Z & I \\ Z & Z & I \end{matrix} \qquad \begin{matrix} I & Z & Z \\ I & Z & Z \\ I & Z & Z \end{matrix}$$

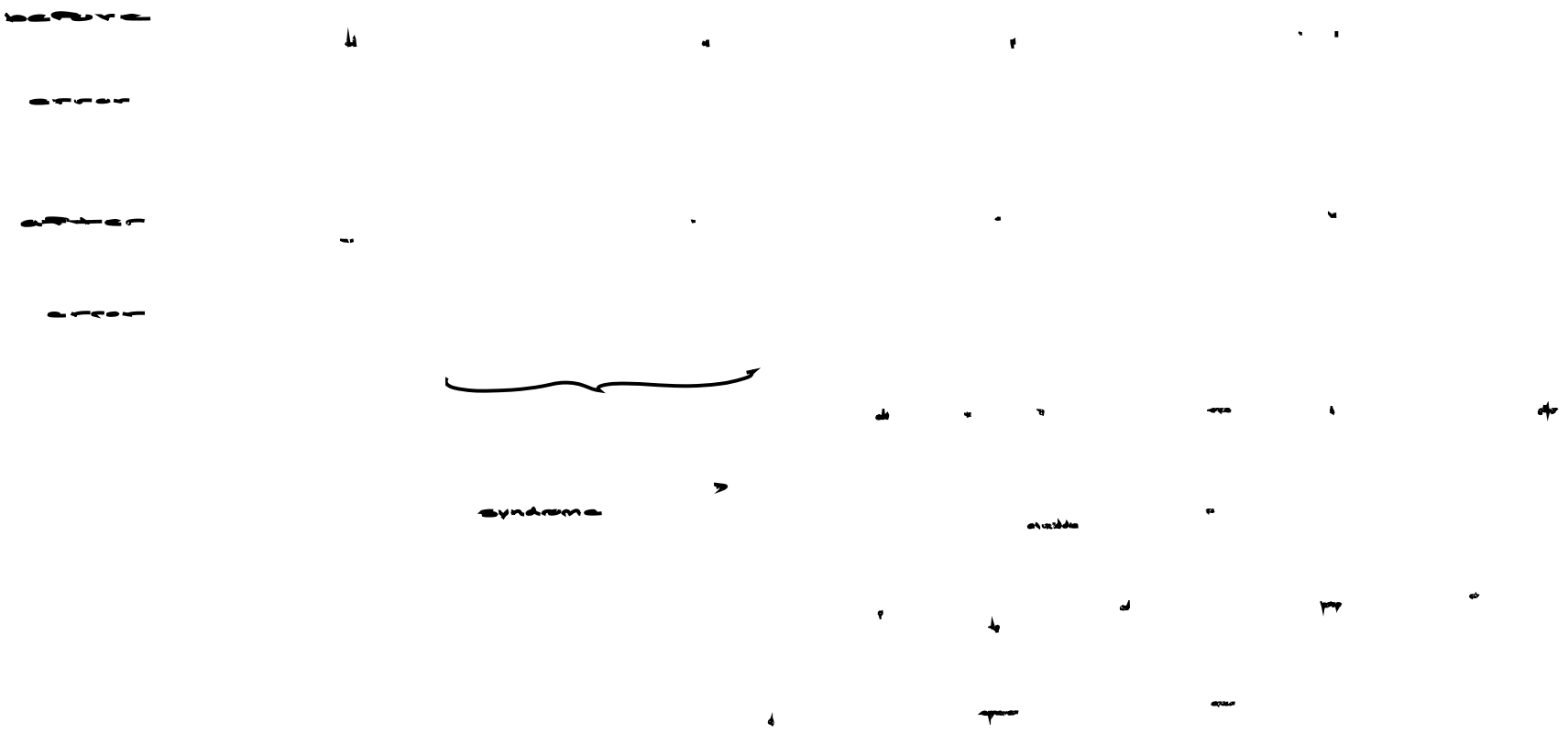
$$\bar{X}_5 = \begin{matrix} X & X & X \\ X & X & X \\ X & X & X \end{matrix} \qquad \bar{Z}_5 = \begin{matrix} Z & Z & Z \\ Z & Z & Z \\ Z & Z & Z \end{matrix}$$

Shor Redux Errors

$$\begin{matrix} I & I & I \\ I & Z & I \\ I & I & I \end{matrix}$$

error

$$\begin{matrix} X & X & X \\ X & X & X \\ I & I & I \end{matrix} \qquad \begin{matrix} I & I & I \\ X & X & X \\ X & X & X \end{matrix} \qquad \begin{matrix} Z & Z & I \\ Z & Z & I \\ Z & Z & I \end{matrix} \qquad \begin{matrix} I & Z & Z \\ I & Z & Z \\ I & Z & Z \end{matrix}$$



$$\begin{matrix} I & I & I \\ I & Z & I \\ I & I & I \end{matrix}$$

$$\begin{matrix} X & X & X \\ X & X & X \\ I & I & I \end{matrix} \qquad \begin{matrix} I & I & I \\ X & X & X \\ X & X & X \end{matrix} \qquad \begin{matrix} Z & Z & I \\ Z & Z & I \\ Z & Z & I \end{matrix} \qquad \begin{matrix} I & Z & Z \\ I & Z & Z \\ I & Z & Z \end{matrix}$$

$$\begin{matrix} I & I & I \\ Z & I & I \\ I & I & I \end{matrix} \qquad \begin{matrix} I & I & I \\ I & Z & I \\ I & I & I \end{matrix} \qquad \begin{matrix} I & I & I \\ I & I & Z \\ I & I & I \end{matrix}$$

$$\begin{matrix} I & I & I \\ Z & Z & I \\ I & I & I \end{matrix} \qquad \begin{matrix} I & I & I \\ I & I & I \\ I & I & I \end{matrix} \qquad \begin{matrix} I & I & I \\ I & Z & Z \\ I & I & I \end{matrix}$$

Subsystem Code

Stabilizer generators:

$$\begin{matrix} Z & Z & I & I & Z & Z \\ Z & Z & I & I & Z & Z \\ Z & Z & I & I & Z & Z \end{matrix}$$

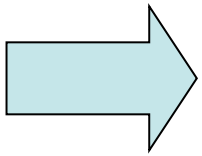
plus encoded Z operators

$$\bar{Z}_1 = \begin{matrix} Z & Z & I \\ I & I & I \\ I & I & I \end{matrix} \quad \bar{Z}_2 = \begin{matrix} I & Z & Z \\ I & I & I \\ I & I & I \end{matrix} \quad \bar{Z}_3 = \begin{matrix} I & I & I \\ I & I & I \\ Z & Z & I \end{matrix} \quad \bar{Z}_4 = \begin{matrix} I & I & I \\ I & I & I \\ I & Z & Z \end{matrix}$$

generate all even number of Z's per row

Measuring +1,-1 eigenvalue of

$$\begin{matrix} X & X & X & I & I & I \\ X & X & X & X & X & X \\ I & I & I & X & X & X \end{matrix}$$



Fix modulo this group i.e. do nothing to 5th encoded qubit!

[Bacon, Phys. Rev A, 73, 012340, 2006]

Generalized Shor Subsystem Codes

Two classical codes: C1=[n1,k1,d1] and C2=[n2,k2,d2]

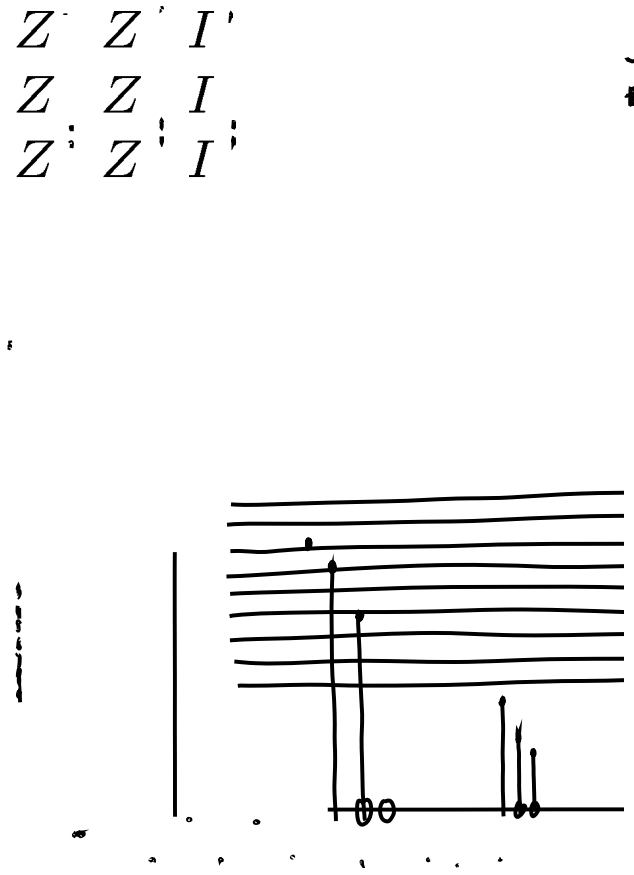
Quantum Shor subspace code [[n1n2,k1k2,min(d1,d2)]]
Number of stabilizers = (n1-k1)n2+(n2-k2)

Quantum Shor subsystem code [[n1n2,k1k2,min(d1,d2)]]
Number of stabilizers= (n1-k1)k2+(n2-k2)k1



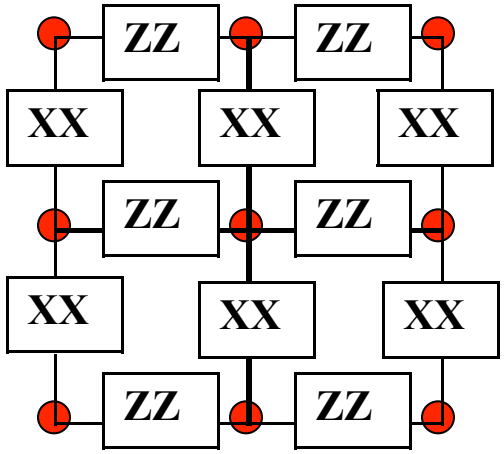
[Bacon, to be posted next wekk, 2006]

Panos Aliferis/Andrew Cross 2006



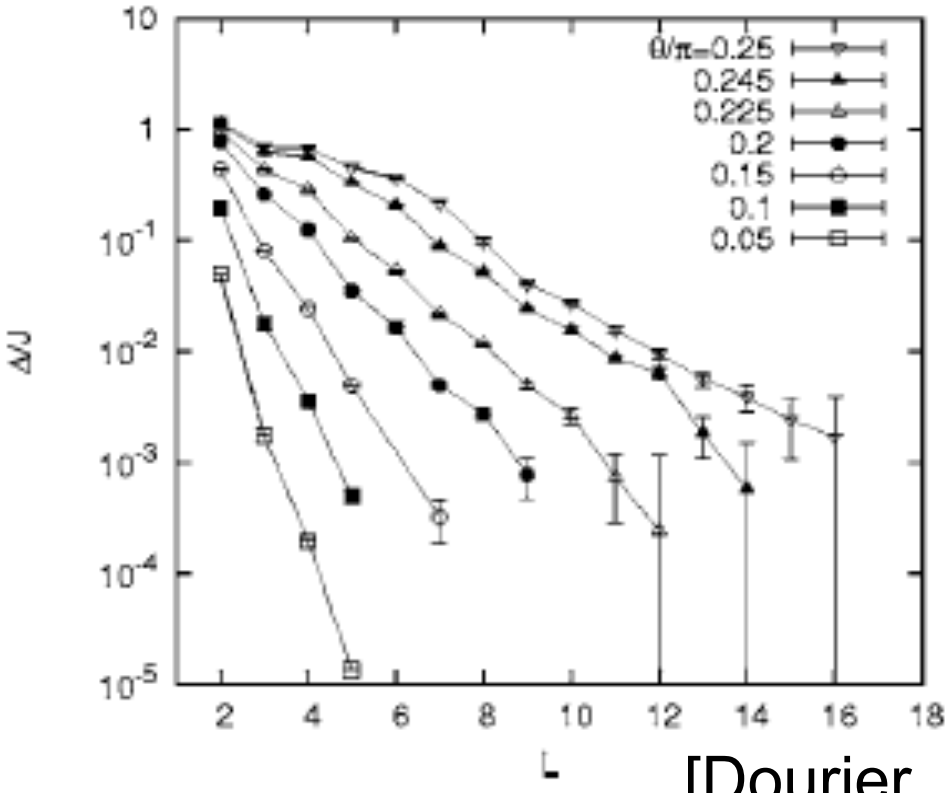
Quantum Compass Model

$$H = -J_X \sum_{ij} X_{i,j} X_{i+1,j} - J_Z \sum_{ij} Z_{i,j} Z_{i,j+1}$$



Ground state is a new kind of quantum error correcting code:
quantum error correcting subsystem

Problems:

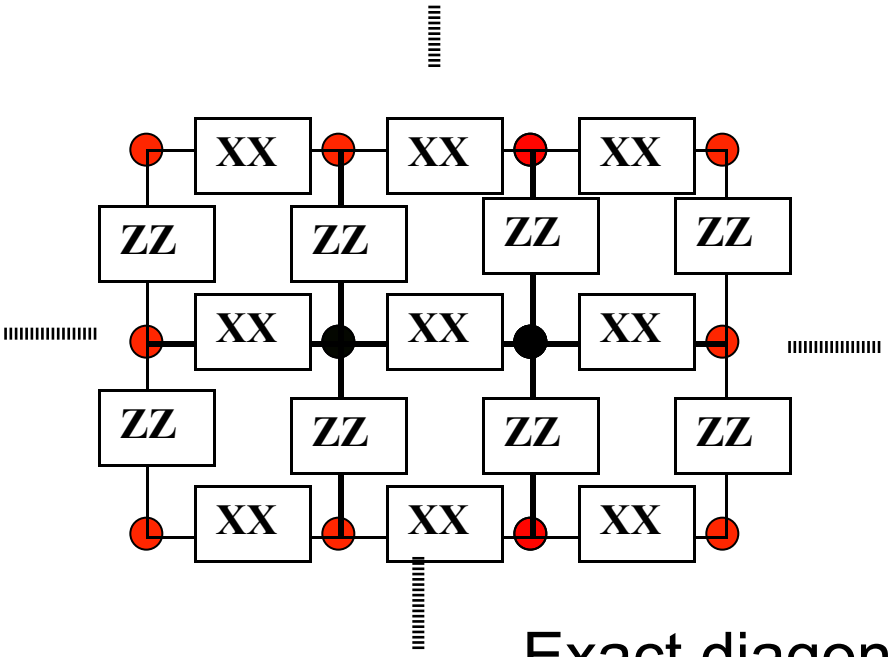


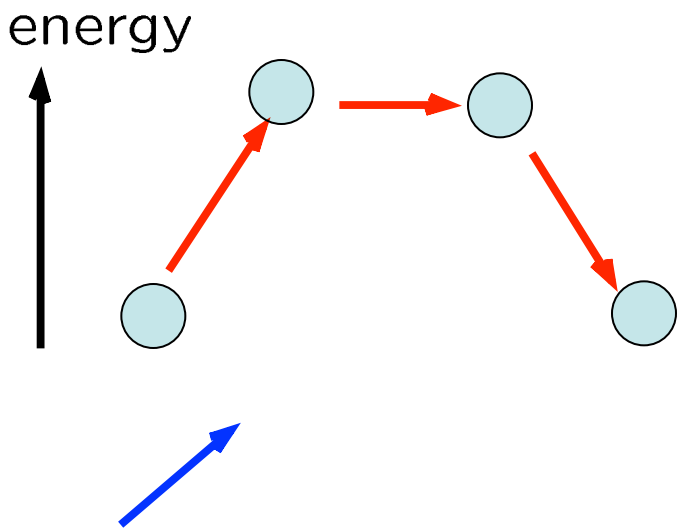
[Dourier, Becca, Mila 05]

$$H = -J \sum_{i,j=1}^{n} \cos(\theta) X_{i,j} X_{i+1,j} + \sin(\theta) Z_{i,j} Z_{i,j+1}$$

no gap in the thermodynamic limit

Problems, Continued

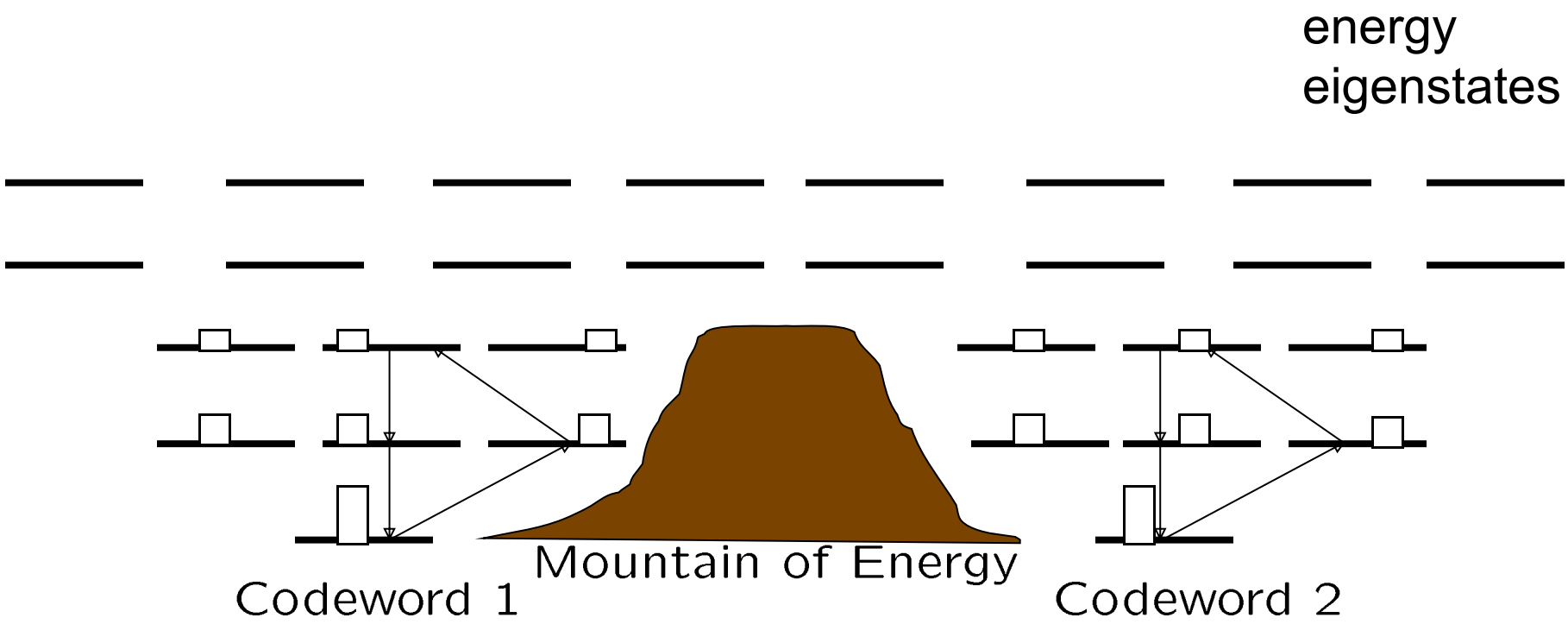




Exact diagonalization 3x3, 4x4, 4x5

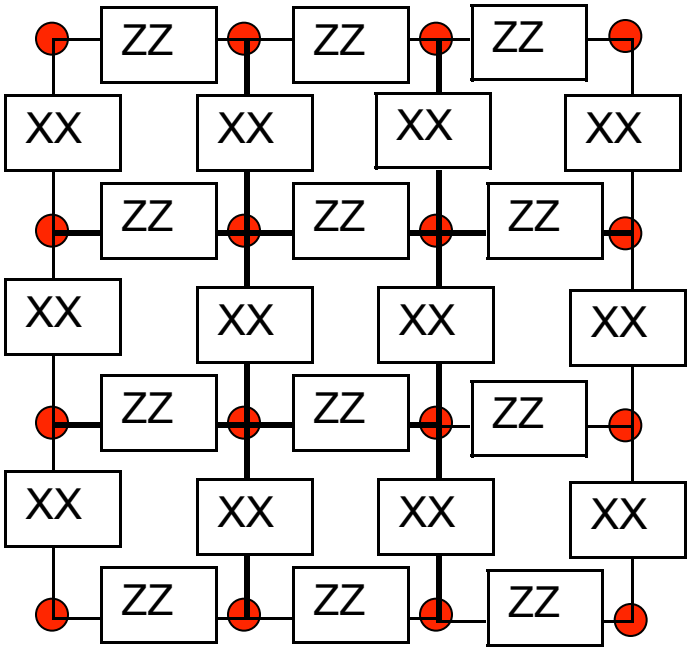
Even at finite size, not “self-correcting”

Self-Correction



Toward Self-Correction

n by *n* square lattice

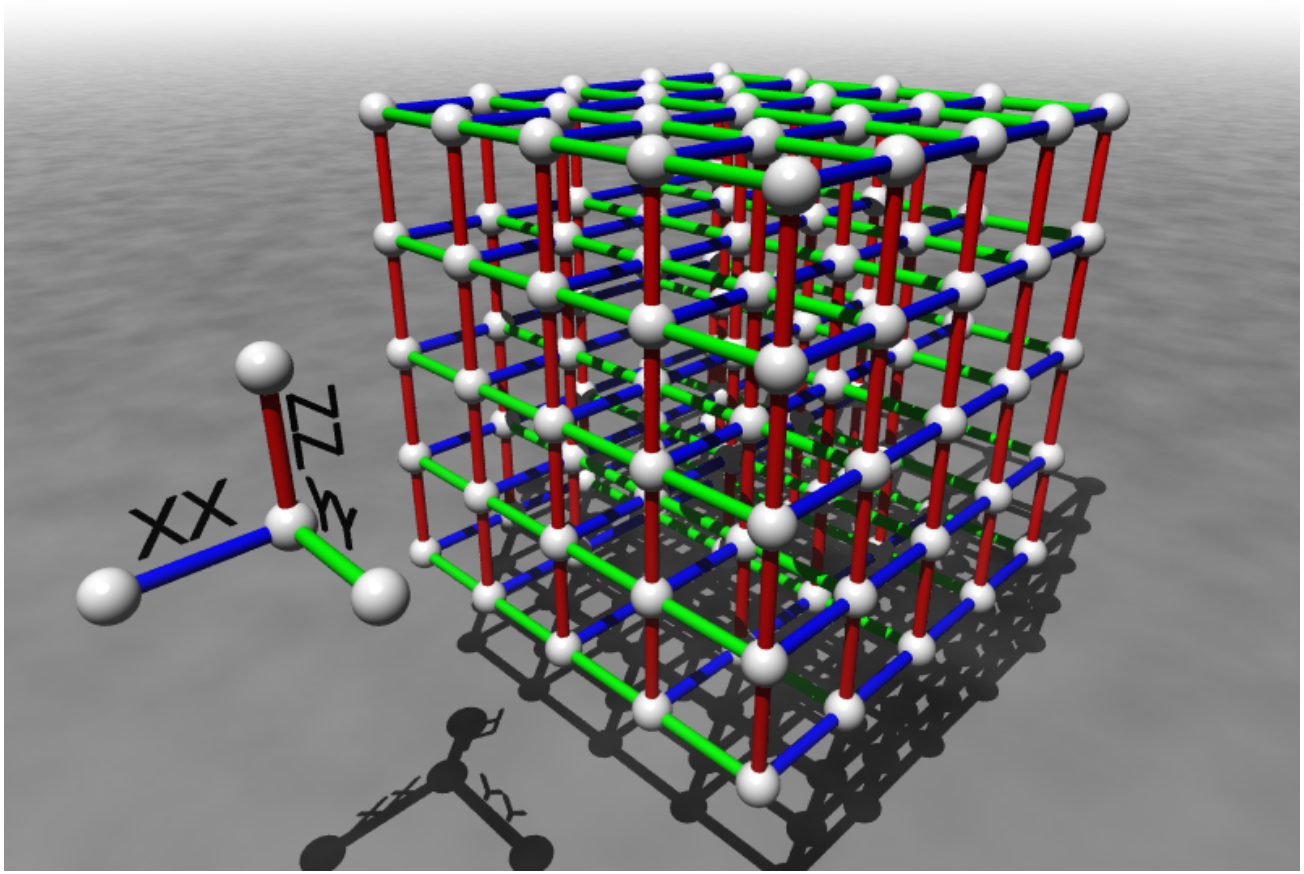


$$H = -\sum_{i,j=1}^{n}\sum_{k=1}^{n-1} J_X(k) X_{i,j} X_{i+k,j} + J_Z(k) Z_{i,j} Z_{i,j+k}$$

Exact diagonalization of 4x4 model, shows that with constant couplings, this model is self-correcting!

Into The Third Dimension

$$H = -J_X \sum_{ijk} X_{i,j,k} X_{i+1,j,k} - J_Y \sum_{ijk} Y_{i,j,k} Y_{i,j+1,k} - J_Z \sum_{ijk} Z_{i,j,k} Z_{i,j,k+1}$$



A Different Path



Potential for robust substrates for quantum information?

